

Lab Title: Week 11 Lab1

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Version: Linux Kali 6.18.12+kali-amd64.

Executive Summary

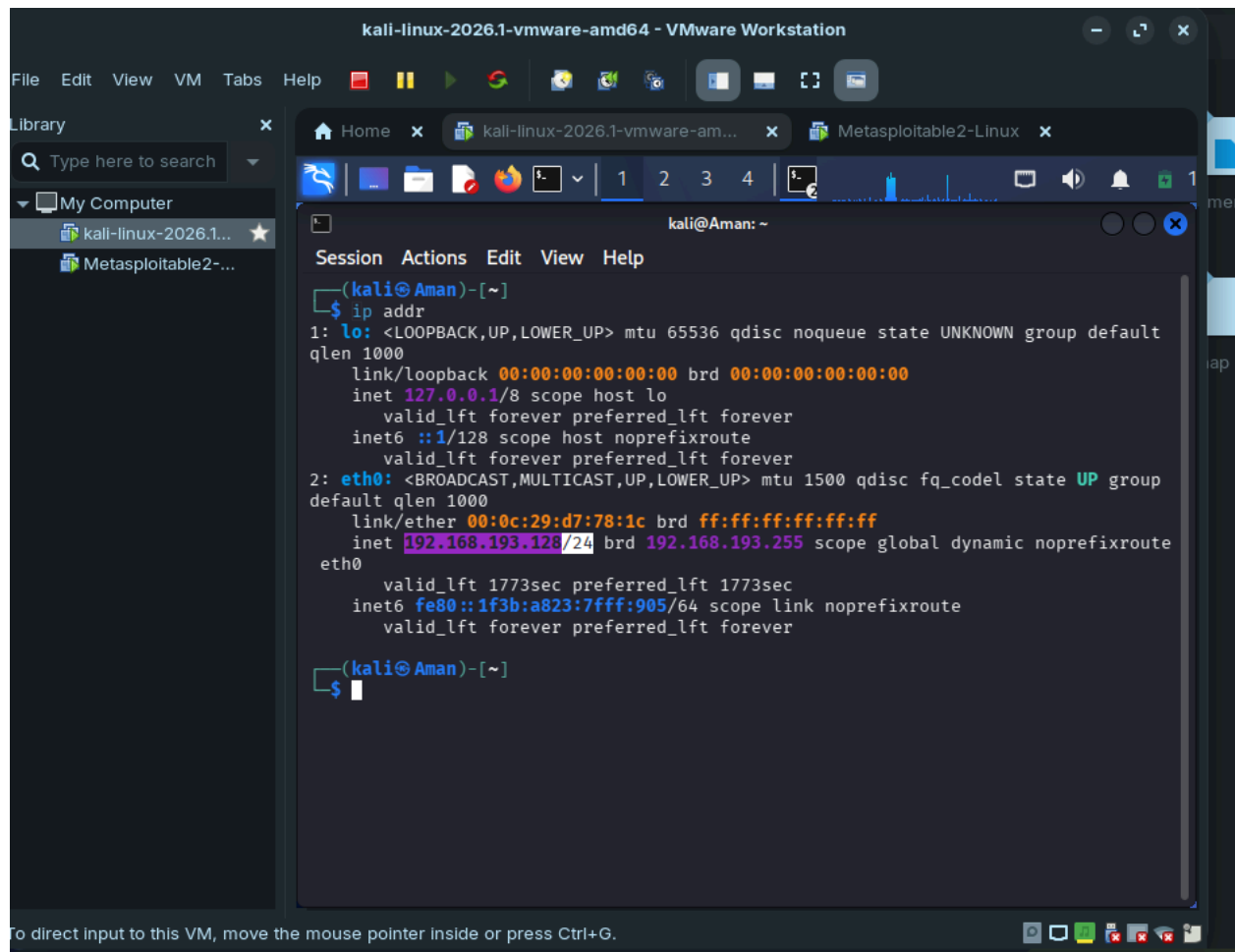
The main focus of this lab is to Understand Nmap Output, Core Nmap Scans, Safe Nmap NSE Enumeration, WhatWeb Fingerprinting and Manual HTTP Confirmation.

Lab Objective

The main objective of this lab is to Understand Network and service reconnaissance.

Methodology

Step 1



The screenshot shows a VMware Workstation window titled "kali-linux-2026.1-vmware-amd64 - VMware Workstation". Inside the VM, a terminal window is open with the prompt "kali@Aman: ~". The user has entered the command "ip addr", and the output is displayed. The output shows details for the loopback interface "lo" and the ethernet interface "eth0". The IP address for "eth0" is 192.168.193.126, and the MAC address is 00:0c:29:d7:78:1c. The terminal output is as follows:

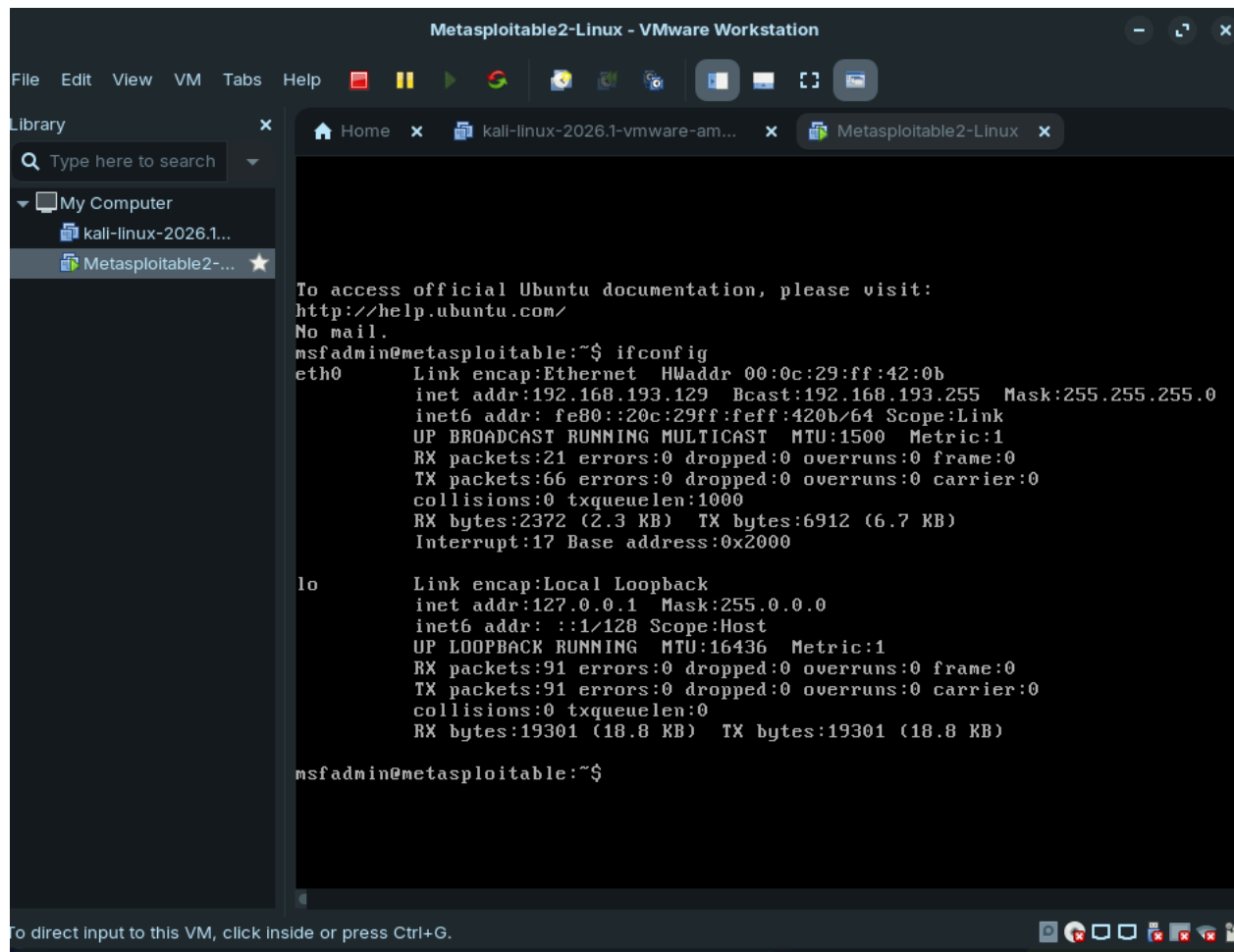
```
(kali@Aman)-[~]
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group
default qlen 1000
    link/ether 00:0c:29:d7:78:1c brd ff:ff:ff:ff:ff:ff
    inet 192.168.193.126/24 brd 192.168.193.255 scope global dynamic noprefixroute
        eth0
        valid_lft 1773sec preferred_lft 1773sec
    inet6 fe80::1f3b:a823:7fff:905/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(kali@Aman)-[~]
$
```

Fig.1A

- I used the command “ip addr” to find my Kali IP address, 192.168.56.102, and my interface eth0, as shown in Fig. 1A above.

Step 2: Find Metasploitable 2 IP



The screenshot shows a VMware Workstation window titled "Metasploitable2-Linux - VMware Workstation". The interface includes a menu bar (File, Edit, View, VM, Tabs, Help), a toolbar with various icons, and a left sidebar with a "Library" section. The "Library" section shows "My Computer" with two items: "kali-linux-2026.1..." and "Metasploitable2-..." (which is selected and marked with a star). The main window displays a terminal window with the following text:

```
To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:ff:42:0b
          inet addr:192.168.193.129  Bcast:192.168.193.255  Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:feff:420b/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:21 errors:0 dropped:0 overruns:0 frame:0
          TX packets:66 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:2372 (2.3 KB)  TX bytes:6912 (6.7 KB)
          Interrupt:17 Base address:0x2000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:91 errors:0 dropped:0 overruns:0 frame:0
          TX packets:91 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:19301 (18.8 KB)  TX bytes:19301 (18.8 KB)

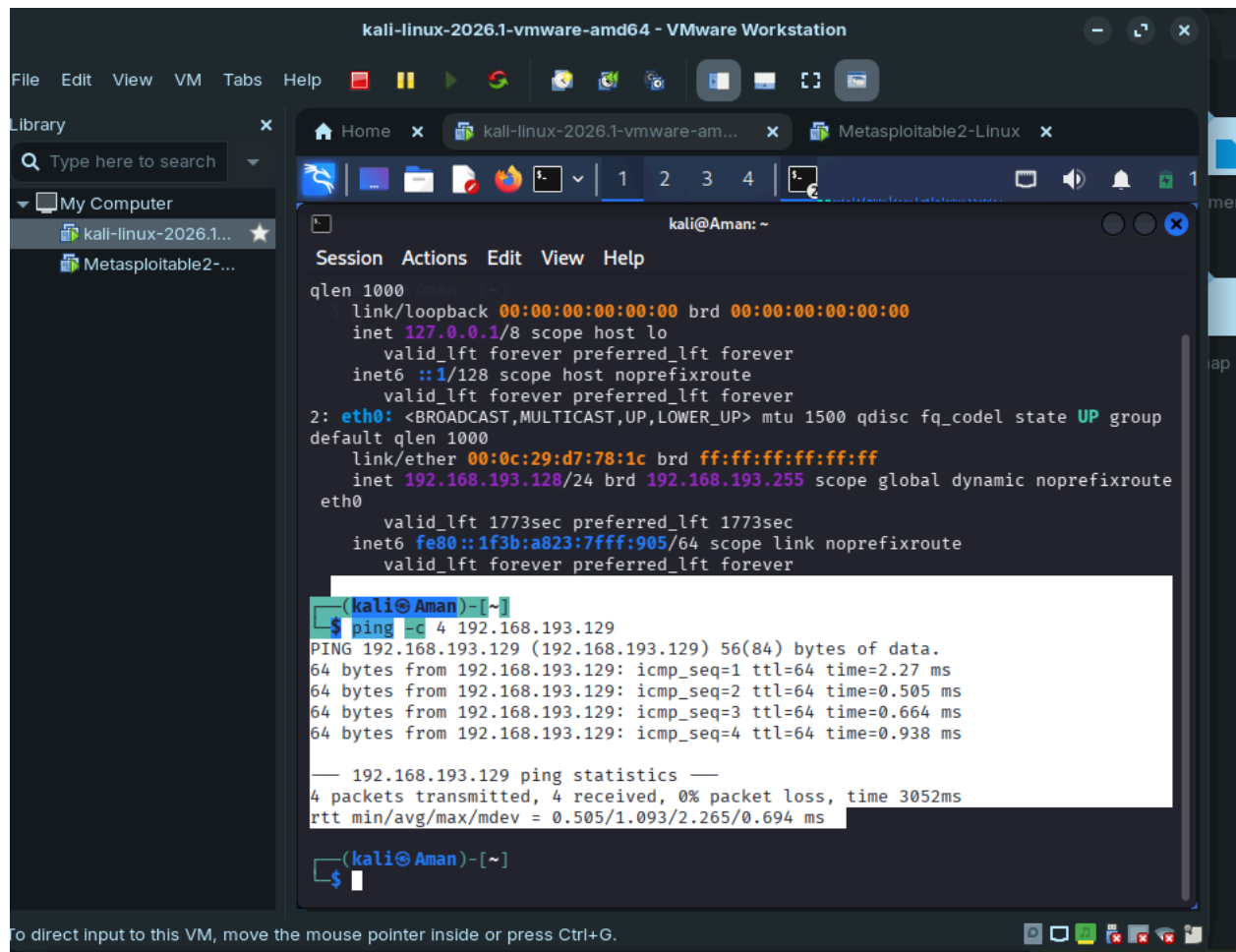
msfadmin@metasploitable:~$
```

At the bottom of the window, there is a status bar that reads "To direct input to this VM, click inside or press Ctrl+G." and a row of icons for window management.

Fig. 1B.

- Using the command “ifconfig”, I found and recorded my Metasploitable interface address (192.168.193.129), as shown in Fig.1B above.

Step 3: Confirm connectivity



```
kali@Aman: ~  
Session Actions Edit View Help  
qlen 1000  
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
inet 127.0.0.1/8 scope host lo  
    valid_lft forever preferred_lft forever  
inet6 ::1/128 scope host noprefixroute  
    valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group  
default qlen 1000  
    link/ether 00:0c:29:d7:78:1c brd ff:ff:ff:ff:ff:ff  
    inet 192.168.193.128/24 brd 192.168.193.255 scope global dynamic noprefixroute  
        eth0  
        valid_lft 1773sec preferred_lft 1773sec  
        inet6 fe80::1f3b:a823:7fff:905/64 scope link noprefixroute  
            valid_lft forever preferred_lft forever  
  
(kali@Aman)-[~]  
$ ping -c 4 192.168.193.129  
PING 192.168.193.129 (192.168.193.129) 56(84) bytes of data:  
64 bytes from 192.168.193.129: icmp_seq=1 ttl=64 time=2.27 ms  
64 bytes from 192.168.193.129: icmp_seq=2 ttl=64 time=0.505 ms  
64 bytes from 192.168.193.129: icmp_seq=3 ttl=64 time=0.664 ms  
64 bytes from 192.168.193.129: icmp_seq=4 ttl=64 time=0.938 ms  
  
— 192.168.193.129 ping statistics —  
4 packets transmitted, 4 received, 0% packet loss, time 3052ms  
rtt min/avg/max/mdev = 0.505/1.093/2.265/0.694 ms  
  
(kali@Aman)-[~]  
$
```

Fig. 1C

- Using the command “ping -c 4 192.168.193.129” i confirmed that my Kali can communicate with the Metasploitable, as shown in Fig. 1C above.

Part 2 - Understand Nmap Output

PORT: Port number and protocol, e.g. 22/tcp.

Open :A service is accepting connections/traffic.

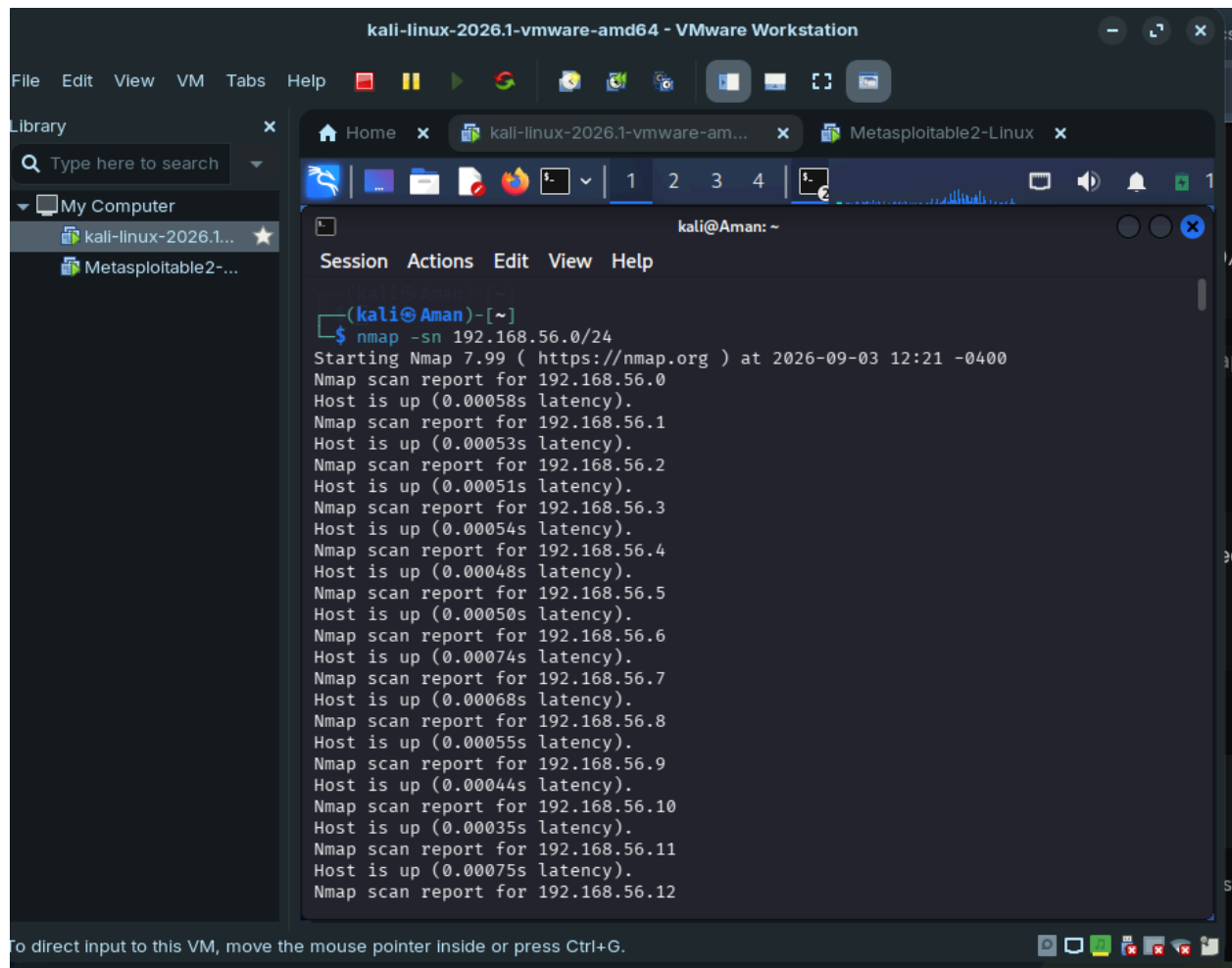
Closed: The host is reachable but nothing is listening on that port.

Filtered: Nmap cannot determine the state because filtering blocks probes/responses.

Service: Nmap service identification.

Version: Product/version information when -sV succeeds

Step 4: Host discovery



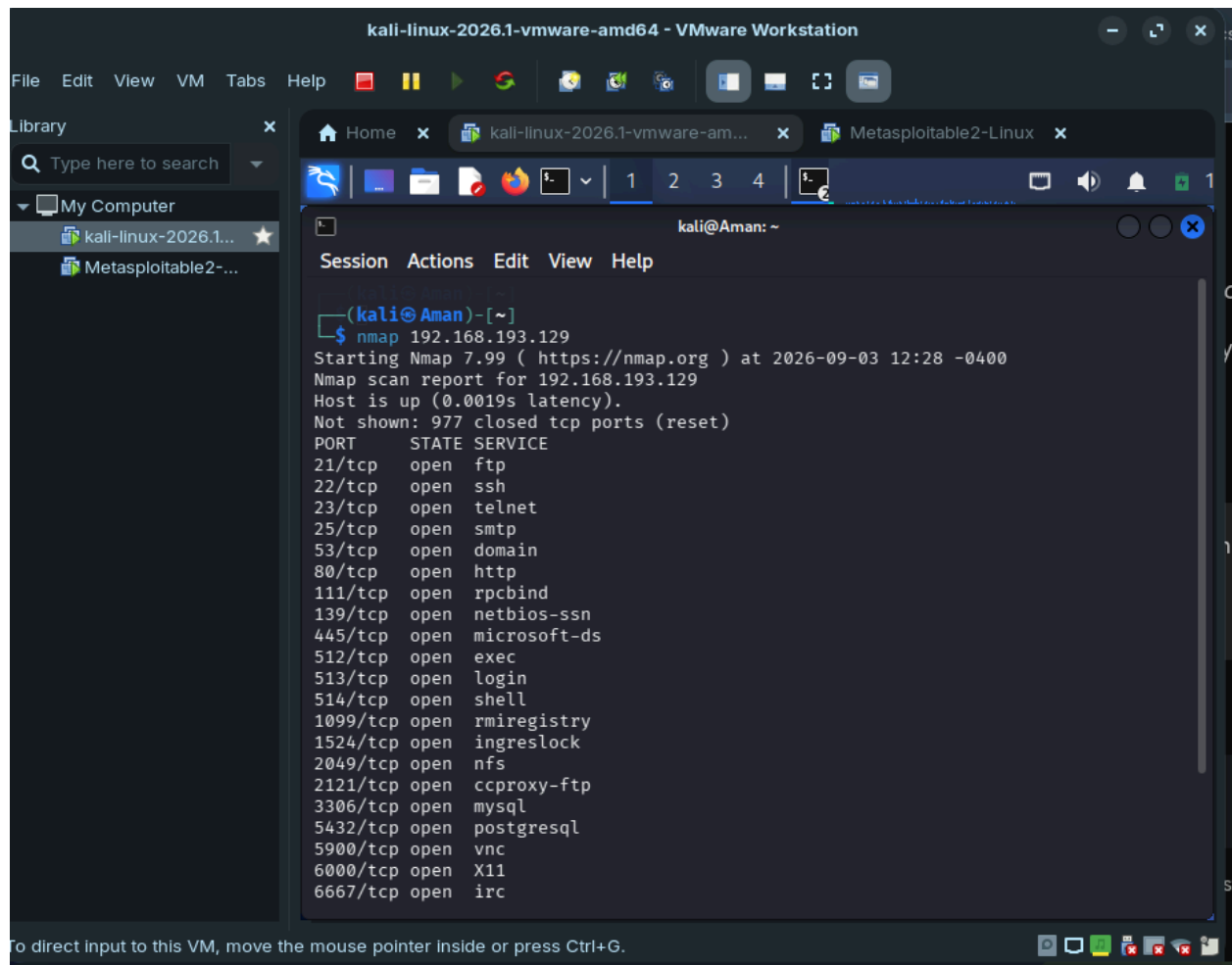
The screenshot shows a VMware Workstation window titled "kali-linux-2026.1-vmware-amd64 - VMware Workstation". Inside, a terminal window titled "kali@Aman: ~" is open. The terminal displays the command `nmap -sn 192.168.56.0/24` and its output, which lists 13 live hosts in the 192.168.56.0/24 network. The hosts are 192.168.56.0 through 192.168.56.12. The terminal also shows the Nmap version (7.99) and the scan time (2026-09-03 12:21 -0400). The VMware interface includes a menu bar (File, Edit, View, VM, Tabs, Help), a toolbar, and a sidebar with a "Library" section showing "My Computer" and two virtual machines: "kali-linux-2026.1..." and "Metasploitable2-...".

```
(kali@Aman)-[~]
$ nmap -sn 192.168.56.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-03 12:21 -0400
Nmap scan report for 192.168.56.0
Host is up (0.00058s latency).
Nmap scan report for 192.168.56.1
Host is up (0.00053s latency).
Nmap scan report for 192.168.56.2
Host is up (0.00051s latency).
Nmap scan report for 192.168.56.3
Host is up (0.00054s latency).
Nmap scan report for 192.168.56.4
Host is up (0.00048s latency).
Nmap scan report for 192.168.56.5
Host is up (0.00050s latency).
Nmap scan report for 192.168.56.6
Host is up (0.00074s latency).
Nmap scan report for 192.168.56.7
Host is up (0.00068s latency).
Nmap scan report for 192.168.56.8
Host is up (0.00055s latency).
Nmap scan report for 192.168.56.9
Host is up (0.00044s latency).
Nmap scan report for 192.168.56.10
Host is up (0.00035s latency).
Nmap scan report for 192.168.56.11
Host is up (0.00075s latency).
Nmap scan report for 192.168.56.12
```

Fig. 2A

- I use the command “nmap -sn 192.168.56.0/24” to discover live host without a normal port scan as shown in Fig. 2A above

Step 5: Default TCP scan



The screenshot shows a Kali Linux virtual machine running in VMware Workstation. A terminal window is open, displaying the results of an Nmap scan. The user has entered the command `nmap 192.168.193.129`. The output shows that the host is up and lists 21 open ports with their respective services.

```
(kali@Aman)~[~]  
$ nmap 192.168.193.129  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-03 12:28 -0400  
Nmap scan report for 192.168.193.129  
Host is up (0.0019s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
111/tcp   open  rpcbind  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
512/tcp   open  exec  
513/tcp   open  login  
514/tcp   open  shell  
1099/tcp  open  rmiregistry  
1524/tcp  open  ingreslock  
2049/tcp  open  nfs  
2121/tcp  open  ccproxy-ftp  
3306/tcp  open  mysql  
5432/tcp  open  postgresql  
5900/tcp  open  vnc  
6000/tcp  open  X11  
6667/tcp  open  irc
```

Fig. 2C

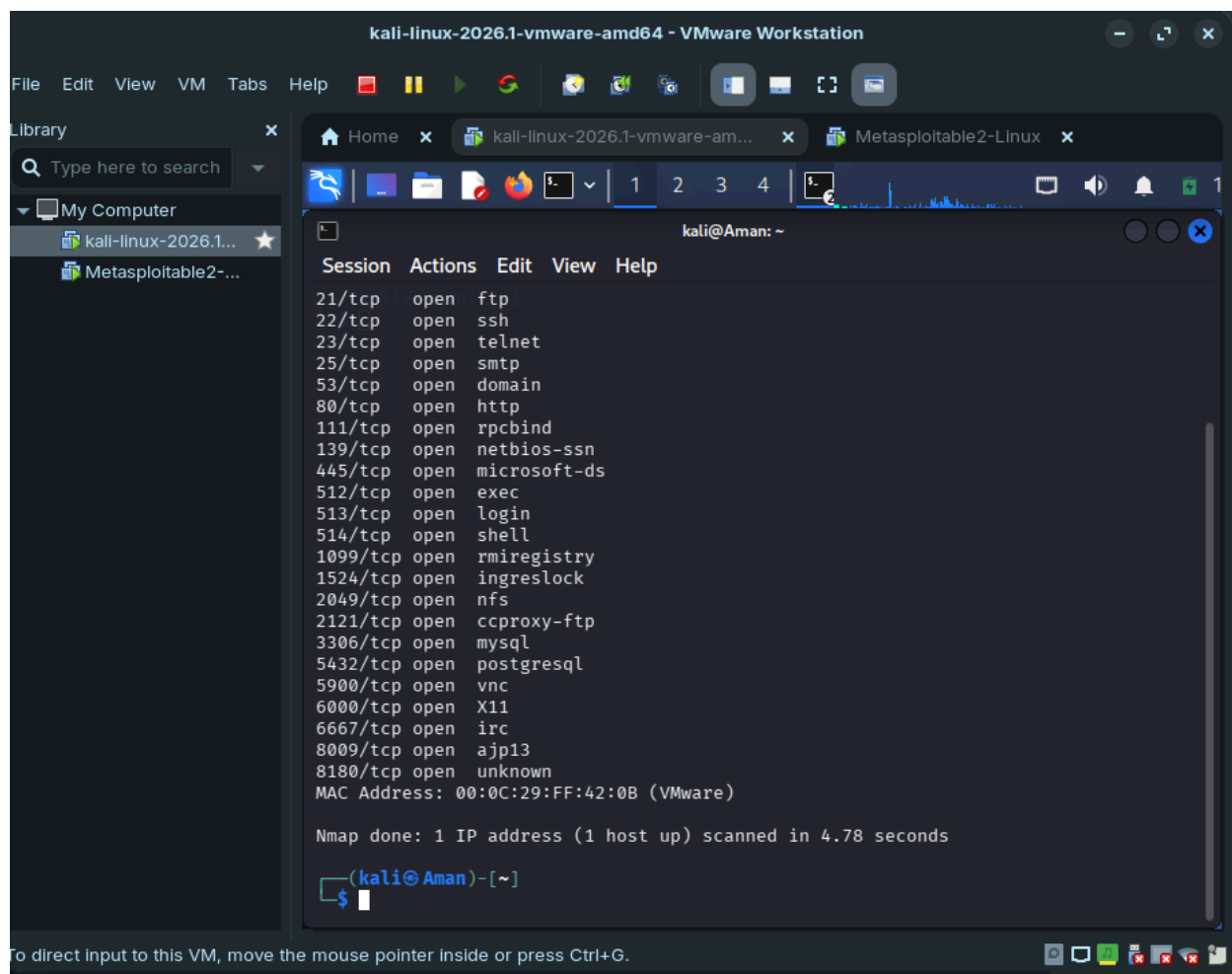
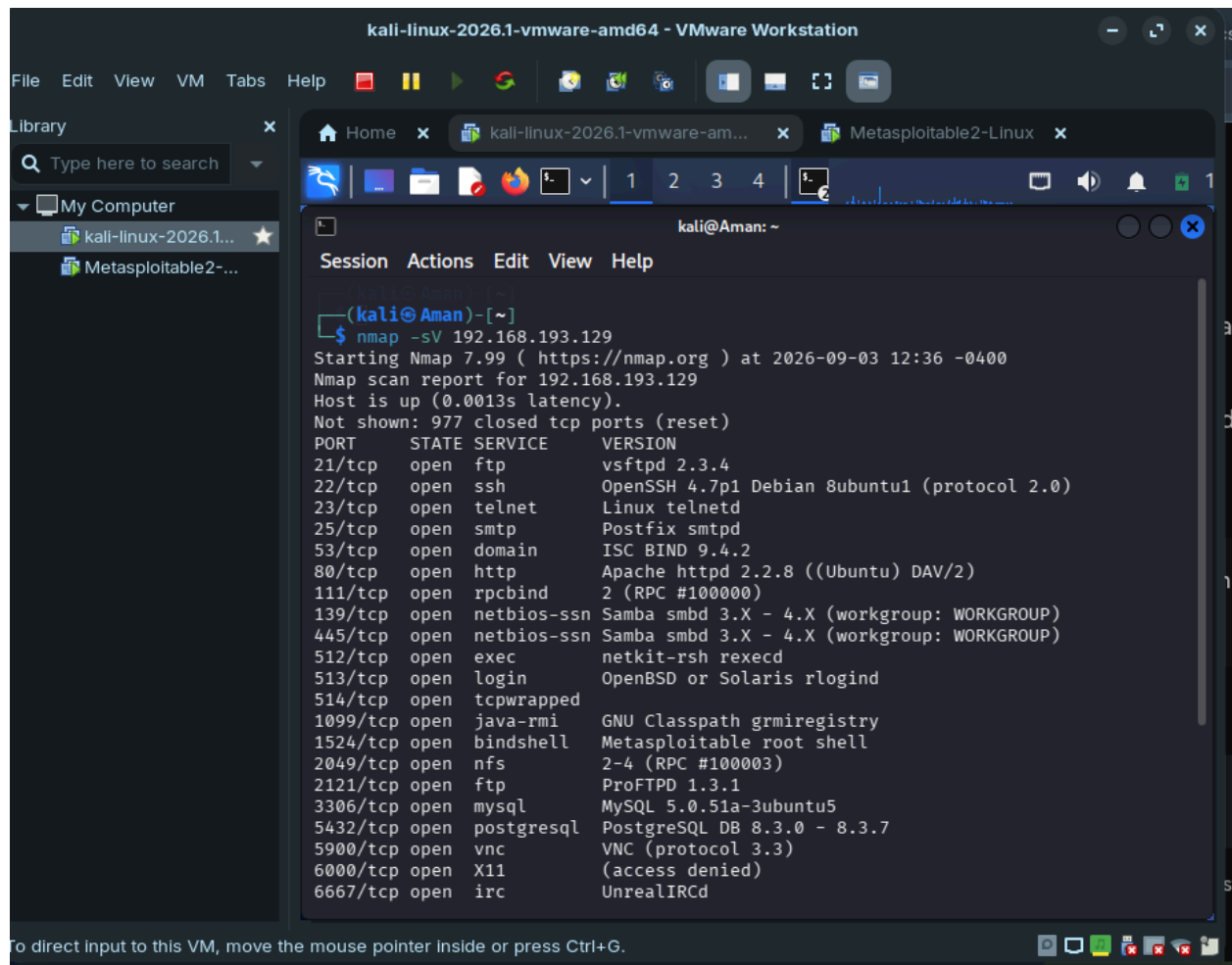


Fig. 2D

- I used the command "nmap 192.168.193.129" to make Nmap scan open TCP ports, and I found 23 TCP port open, as shown in Fig. 2C and Fig. 2D above.

Step 6: Service/version detection



The screenshot shows a Kali Linux terminal window titled "kali@Aman: ~". The terminal displays the output of an Nmap scan performed on the IP address 192.168.193.129. The output includes the Nmap version (7.99), the scan time (2026-09-03 12:36 -0400), and a list of open ports with their corresponding services and versions.

```
(kali@Aman)~[~]  
$ nmap -sV 192.168.193.129  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-03 12:36 -0400  
Nmap scan report for 192.168.193.129  
Host is up (0.0013s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE      VERSION  
21/tcp    open  ftp          vsftpd 2.3.4  
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)  
23/tcp    open  telnet       Linux telnetd  
25/tcp    open  smtp         Postfix smtpd  
53/tcp    open  domain       ISC BIND 9.4.2  
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)  
111/tcp   open  rpcbind      2 (RPC #100000)  
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)  
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)  
512/tcp   open  exec         netkit-rsh rexecd  
513/tcp   open  login        OpenBSD or Solaris rlogind  
514/tcp   open  tcpwrapped  
1099/tcp  open  java-rmi     GNU Classpath grmiregistry  
1524/tcp  open  bindshell    Metasploitable root shell  
2049/tcp  open  nfs          2-4 (RPC #100003)  
2121/tcp  open  ftp          ProFTPD 1.3.1  
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5  
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7  
5900/tcp  open  vnc          VNC (protocol 3.3)  
6000/tcp  open  X11          (access denied)  
6667/tcp  open  irc          UnrealIRCd
```

Fig. 2F.

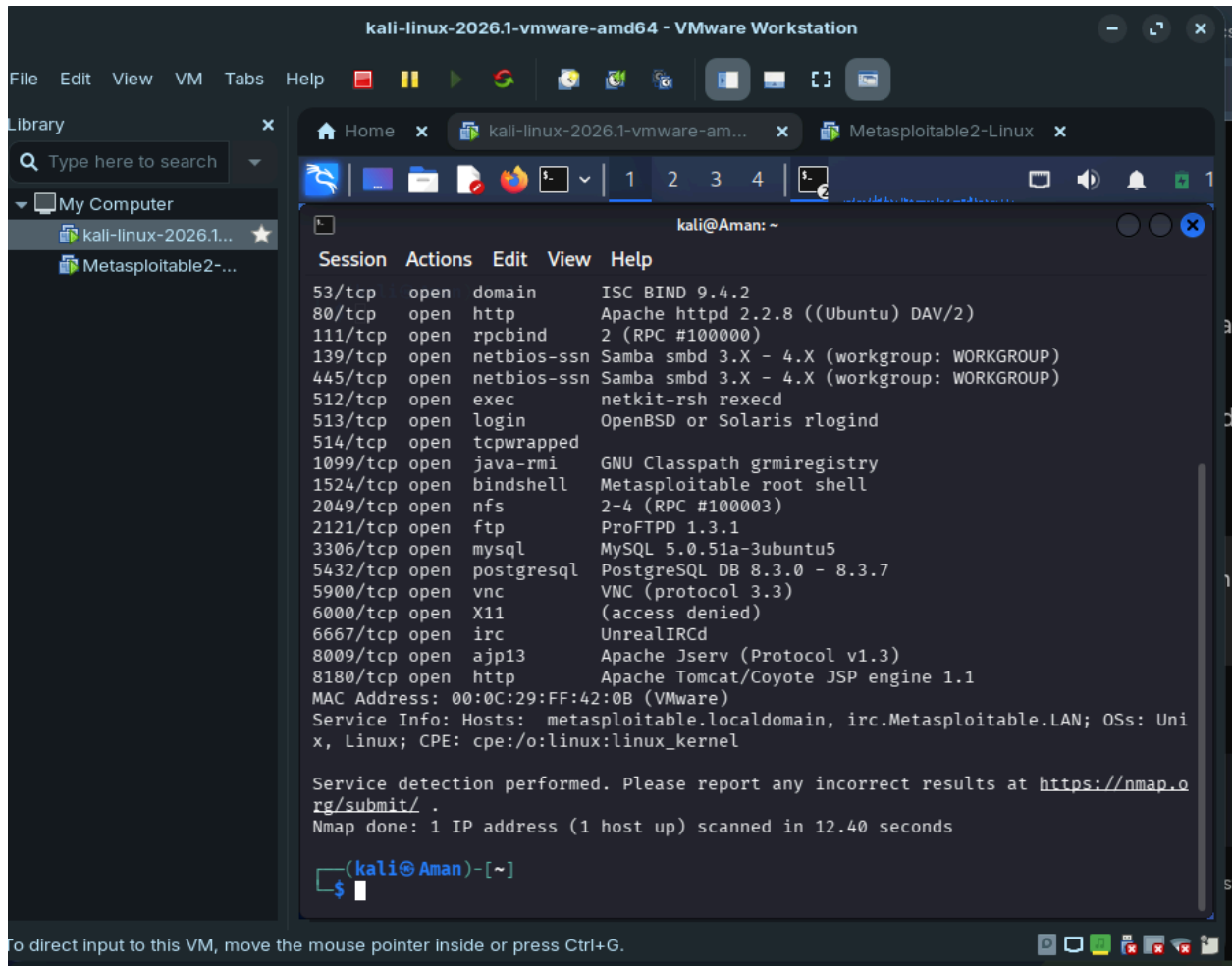


Fig. 2G

- I used the command “nmap -sV 192.168.193.129” to identify software/versions as shown in Fig.2F and Fig.2G above, here is a record of five among what I find:

Port/protocol	Service Name	Identified software and version
21/tcp	ftp	vsftpd 2.3.4
22/tcp	ssh	OpenSSH 4.7p1
80/tcp	http	Apache httpd 2.2.8
1524/tcp	bindshell	Metasploitable root shell
6667/tcp	irc	UnrealIRCd

Step 7: Maximum version intensity

The screenshot shows a Kali Linux virtual machine running in VMware Workstation. The terminal window displays the output of an Nmap scan performed on the IP address 192.168.193.129 using the command `nmap -sV --version-intensity 9 192.168.193.129`. The scan results show a list of open ports and their corresponding services and versions.

```
(kali@Aman)-[~]  
$ nmap -sV --version-intensity 9 192.168.193.129  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-03 12:42 -0400  
Nmap scan report for 192.168.193.129  
Host is up (0.0019s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE          VERSION  
21/tcp    open  ftp              vsftpd 2.3.4  
22/tcp    open  ssh              OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)  
23/tcp    open  telnet           Linux telnetd  
25/tcp    open  smtp             Postfix smtpd  
53/tcp    open  domain          ISC BIND 9.4.2  
80/tcp    open  http             Apache httpd 2.2.8 ((Ubuntu) DAV/2)  
111/tcp   open  rpcbind          2 (RPC #100000)  
139/tcp   open  netbios-ssn      Samba smbd 3.X - 4.X (workgroup: WORKGROUP)  
445/tcp   open  netbios-ssn      Samba smbd 3.X - 4.X (workgroup: WORKGROUP)  
512/tcp   open  exec             netkit-rsh rshcd  
513/tcp   open  login            OpenBSD or Solaris rlogind  
514/tcp   open  tcpwrapped  
1099/tcp  open  java-rmi         GNU Classpath grmiregistry  
1524/tcp  open  bindshell        Metasploitable root shell  
2049/tcp  open  nfs              2-4 (RPC #100003)  
2121/tcp  open  ftp              ProFTPD 1.3.1  
3306/tcp  open  mysql            MySQL 5.0.51a-3ubuntu5  
5432/tcp  open  postgresql       PostgreSQL DB 8.3.0 - 8.3.7  
5900/tcp  open  vnc              VNC (protocol 3.3)  
6000/tcp  open  X11              (access denied)  
6667/tcp  open  irc              UnrealIRCd
```

Fig. 2H

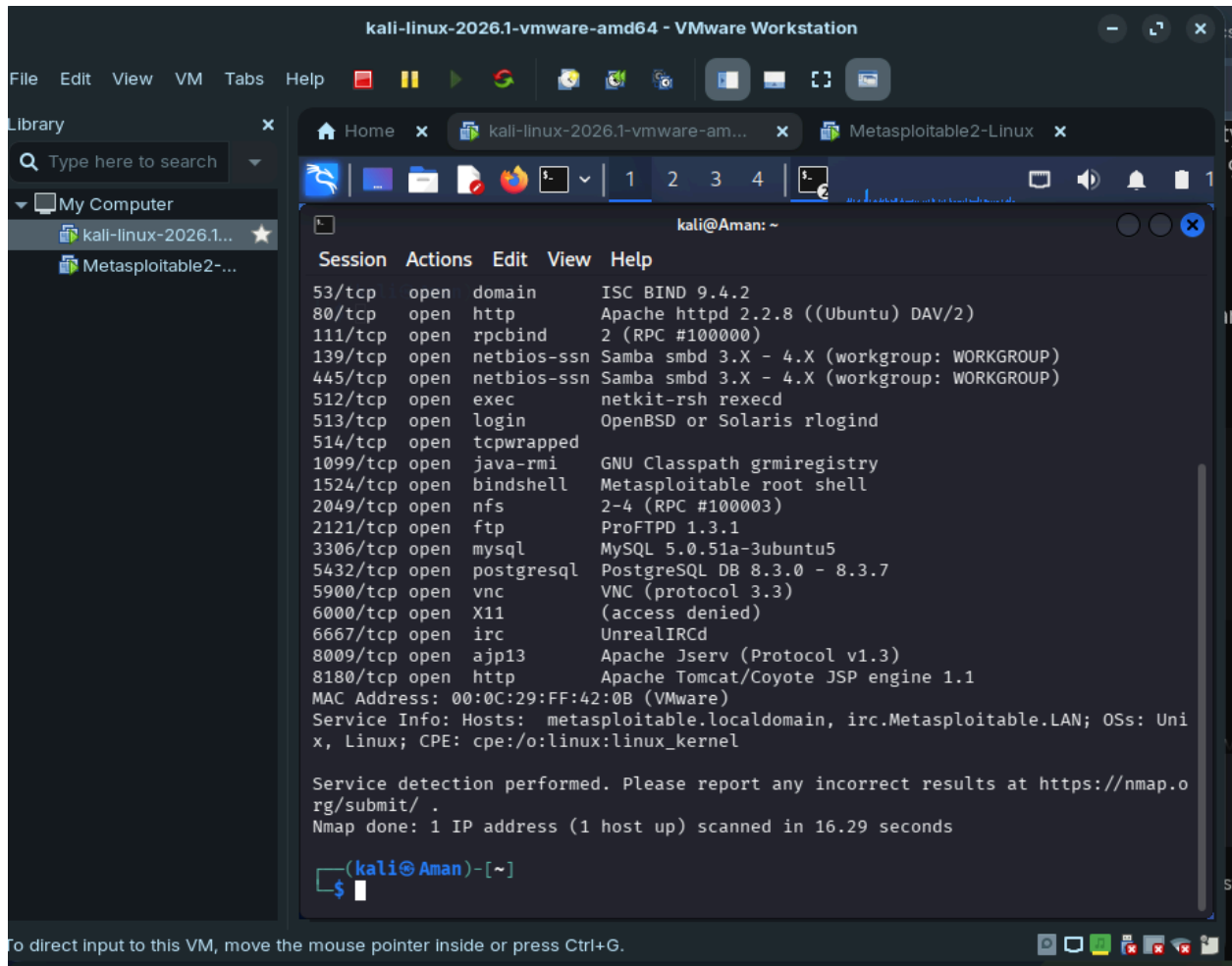
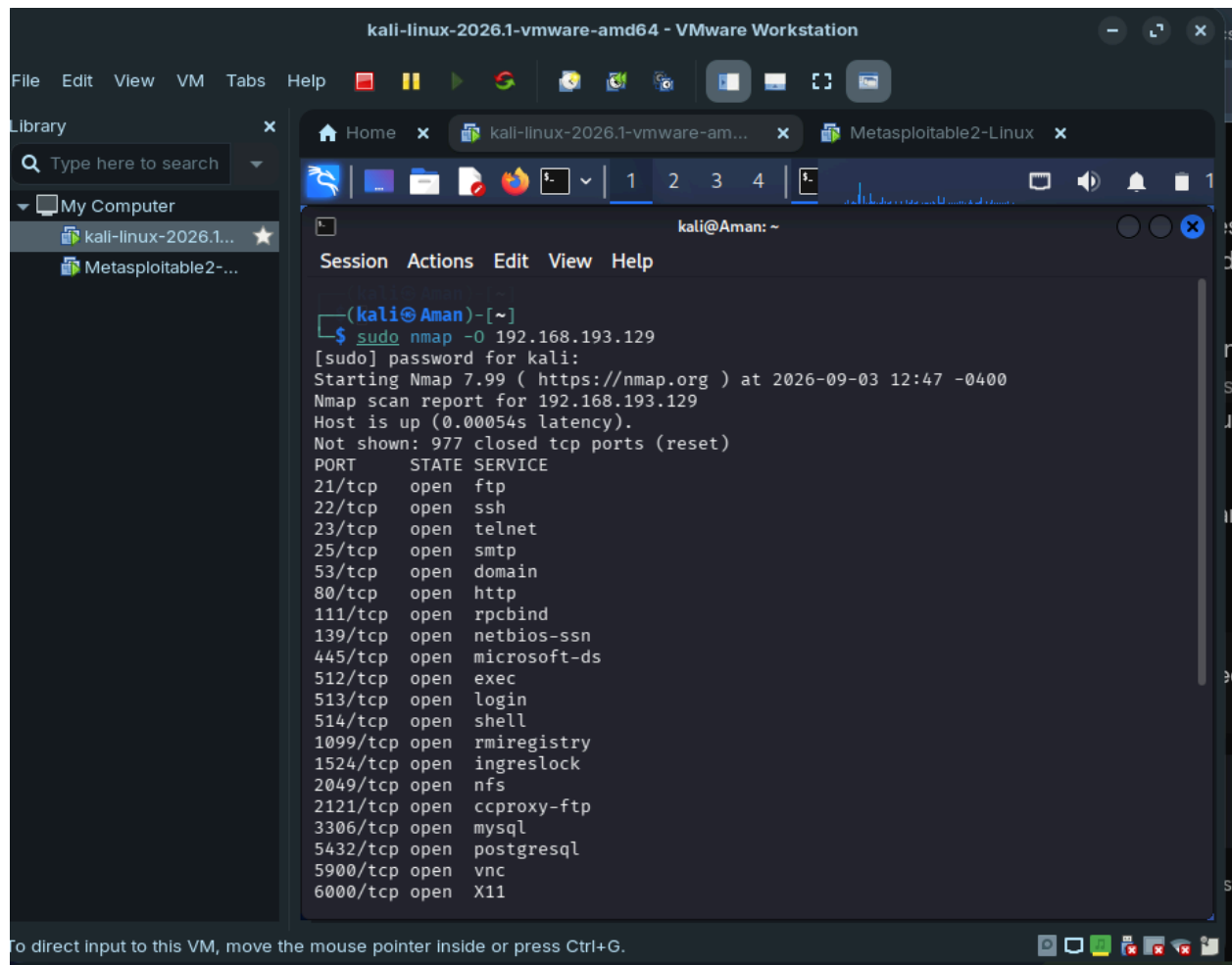


Fig. 21

- Using “nmap -sV --version-intensity 9 <target-ip>” I did a deeper version probing with level nine intensity as shown in Fig. 2H and Fig. 2I. In comparison with normal -sV though the result looks almost similar the deeper probe identifies obscure services that standard probes miss.

Step 8: OS detection



The screenshot shows a VMware Workstation interface with a Kali Linux VM named 'kali-linux-2026.1-vmware-amd64'. The terminal window is open, showing the output of a command. The command is `sudo nmap -O 192.168.193.129`. The output shows the Nmap scan report for 192.168.193.129, indicating that the host is up and listing 21 open ports with their corresponding services.

```
(kali@Aman)-[~]  
$ sudo nmap -O 192.168.193.129  
[sudo] password for kali:  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-03 12:47 -0400  
Nmap scan report for 192.168.193.129  
Host is up (0.00054s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
111/tcp   open  rpcbind  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
512/tcp   open  exec  
513/tcp   open  login  
514/tcp   open  shell  
1099/tcp  open  rmiregistry  
1524/tcp  open  ingreslock  
2049/tcp  open  nfs  
2121/tcp  open  ccproxy-ftp  
3306/tcp  open  mysql  
5432/tcp  open  postgresql  
5900/tcp  open  vnc  
6000/tcp  open  X11
```

Fig. 2J

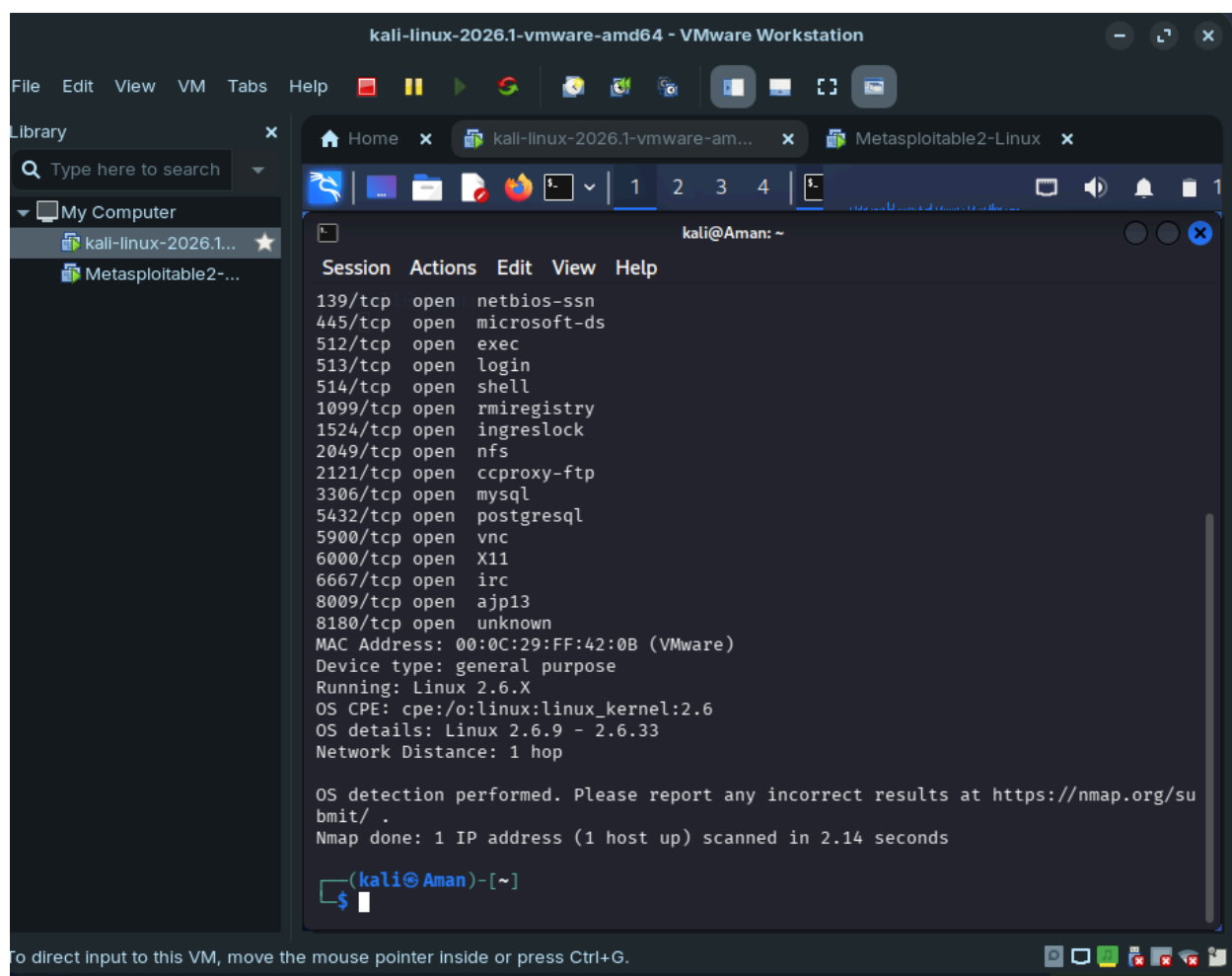


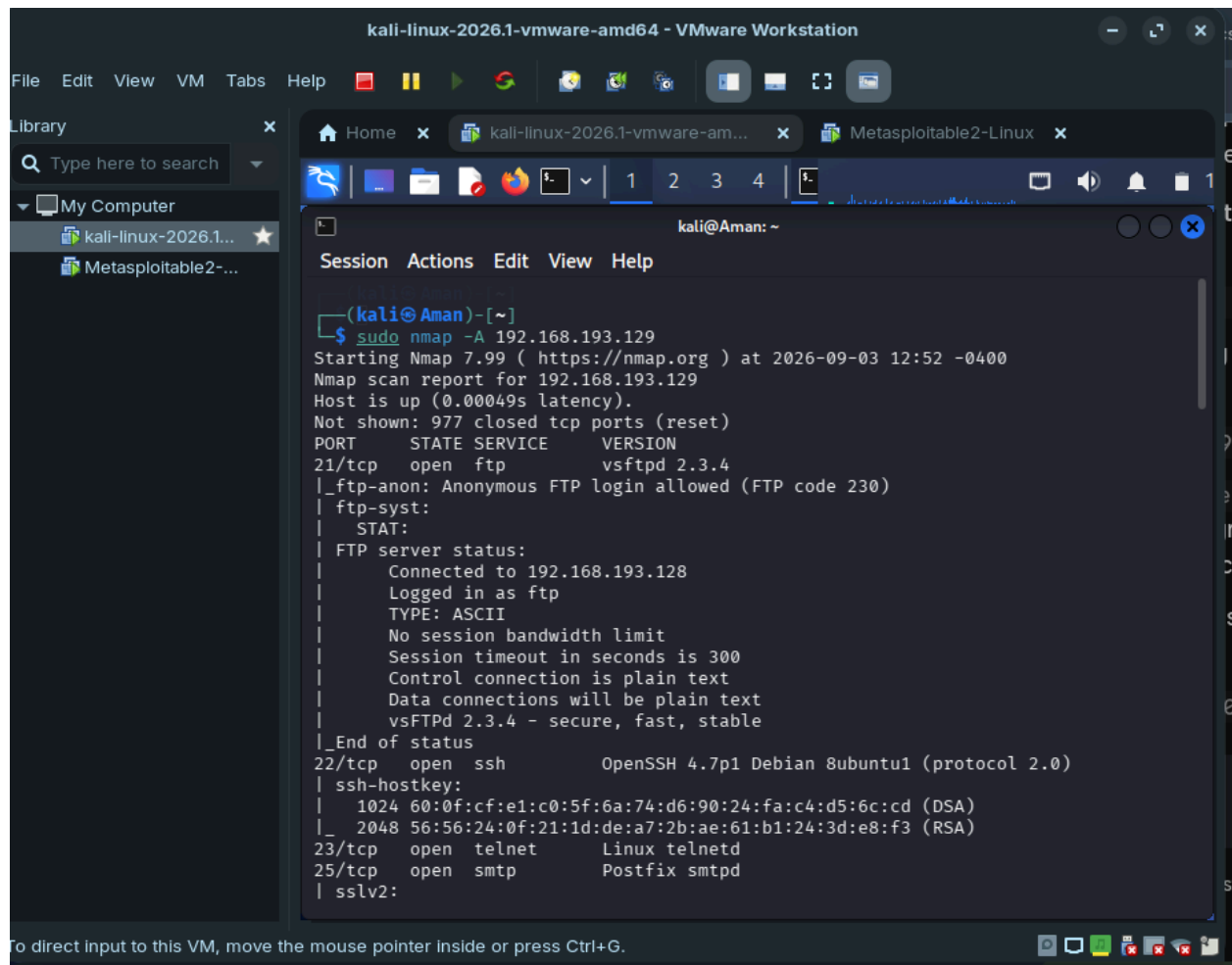
Fig. 2k

- I use the command `sudo nmap -O 192.168.193.129` to detect the target OS as shown in Fig. 2J and Fig. 2k above.

Operating System Estimate

- Target IP: 192.168.193.129
- Device Type: General purpose
- OS Family / Kernel: Linux 2.6.X
- OS Details (Estimate): Linux 2.6.9 – 2.6.33
- OS CPE: `cpe:/o:linux:linux_kernel:2.6`
- Network Distance: 1 hop

Step 9: Aggressive scan



The screenshot shows a Kali Linux virtual machine running in VMware Workstation. The terminal window displays the output of an aggressive Nmap scan performed on the IP address 192.168.193.129. The scan identifies several open ports and services, including FTP, SSH, Telnet, and SMTP. The user is logged in as 'ftp' on the target host.

```
kali@Aman: ~  
$ sudo nmap -A 192.168.193.129  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-03 12:52 -0400  
Nmap scan report for 192.168.193.129  
Host is up (0.00049s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE      VERSION  
21/tcp    open  ftp          vsftpd 2.3.4  
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)  
|_ftp-syst:  
|   STAT:  
|_FTP server status:  
|   Connected to 192.168.193.128  
|   Logged in as ftp  
|   TYPE: ASCII  
|   No session bandwidth limit  
|   Session timeout in seconds is 300  
|   Control connection is plain text  
|   Data connections will be plain text  
|   vsFTPd 2.3.4 - secure, fast, stable  
|_End of status  
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)  
|_ssh-hostkey:  
|   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)  
|   2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)  
23/tcp    open  telnet       Linux telnetd  
25/tcp    open  smtp         Postfix smtpd  
|_sslv2:
```

Fig. 2L

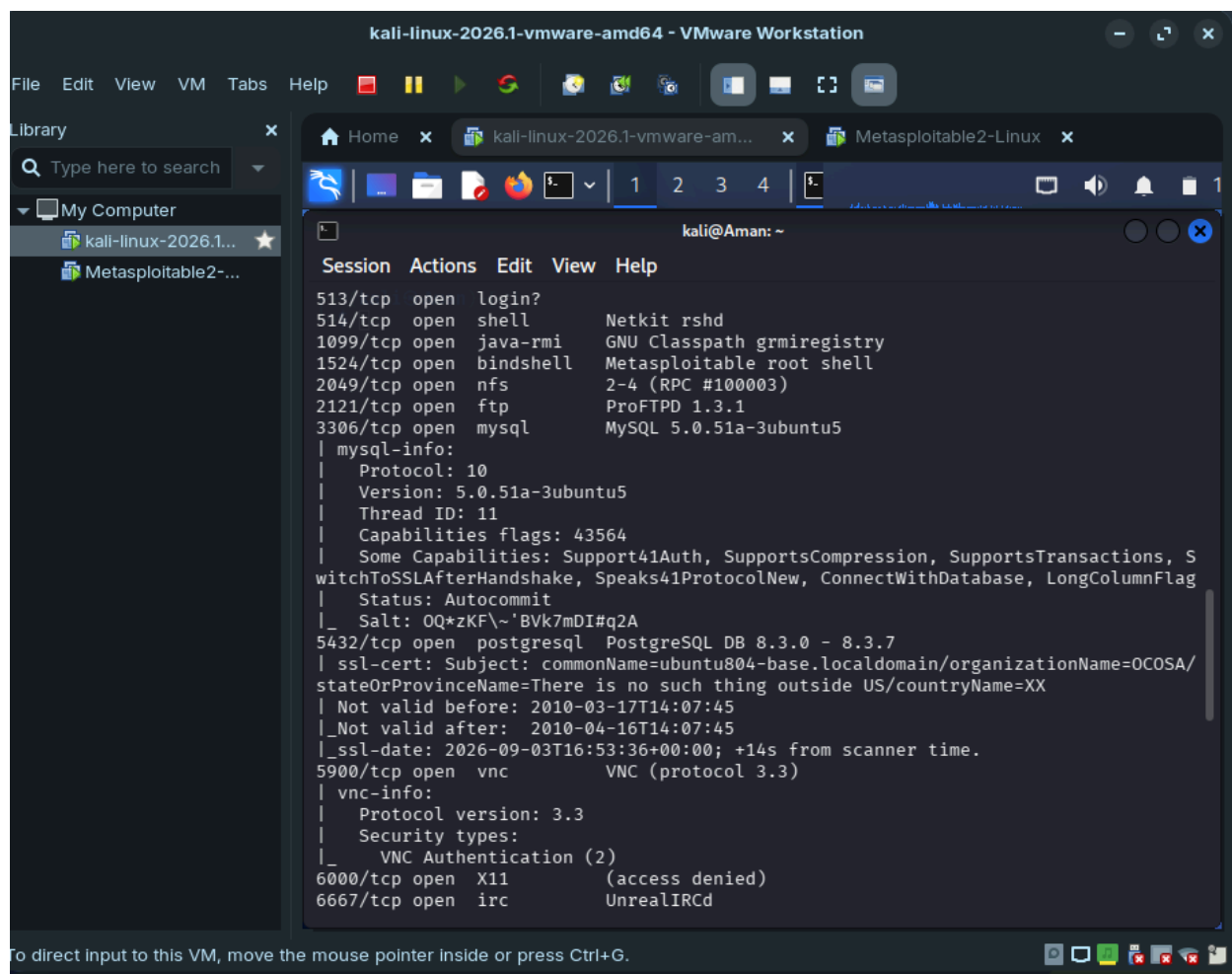


Fig. 2M

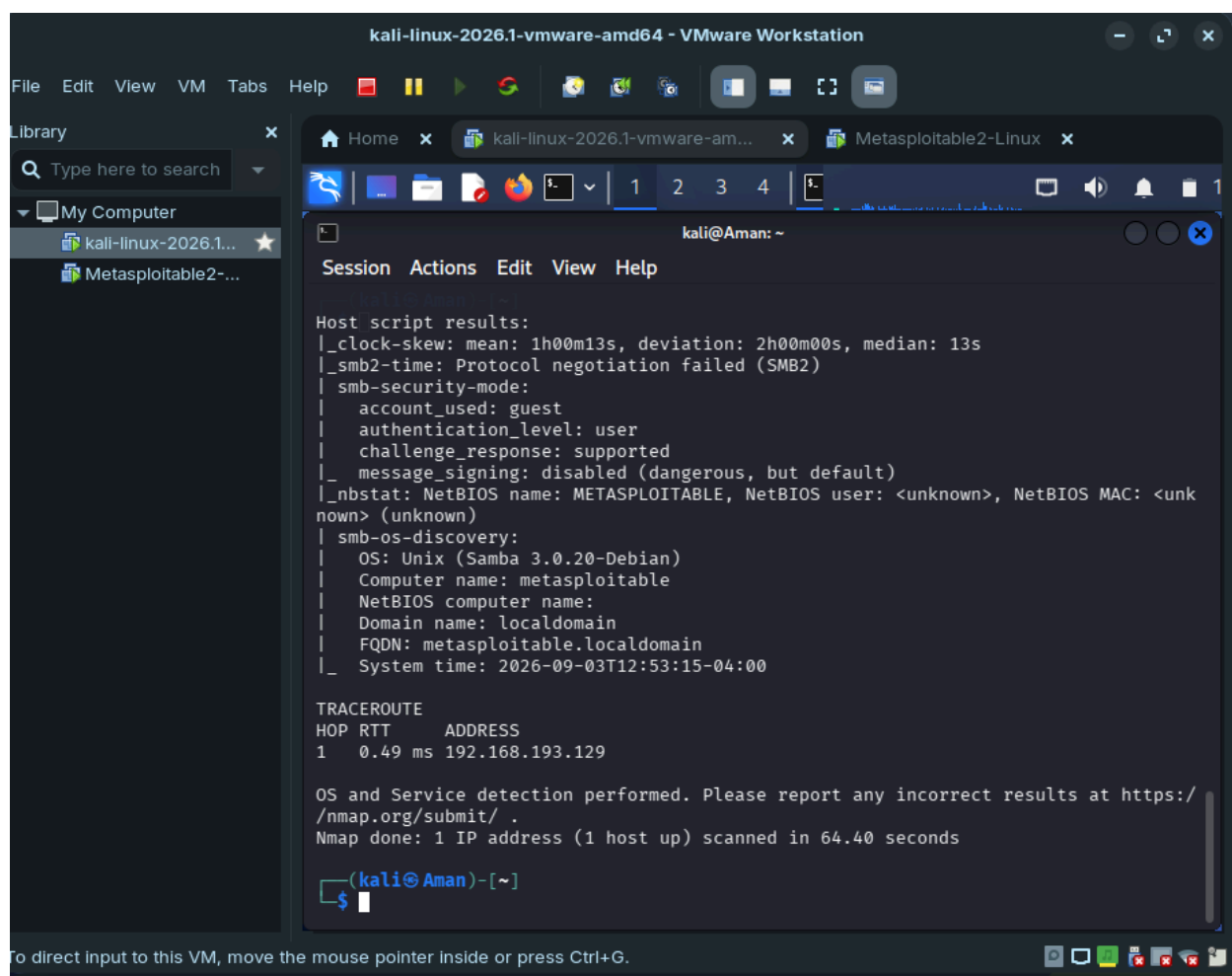
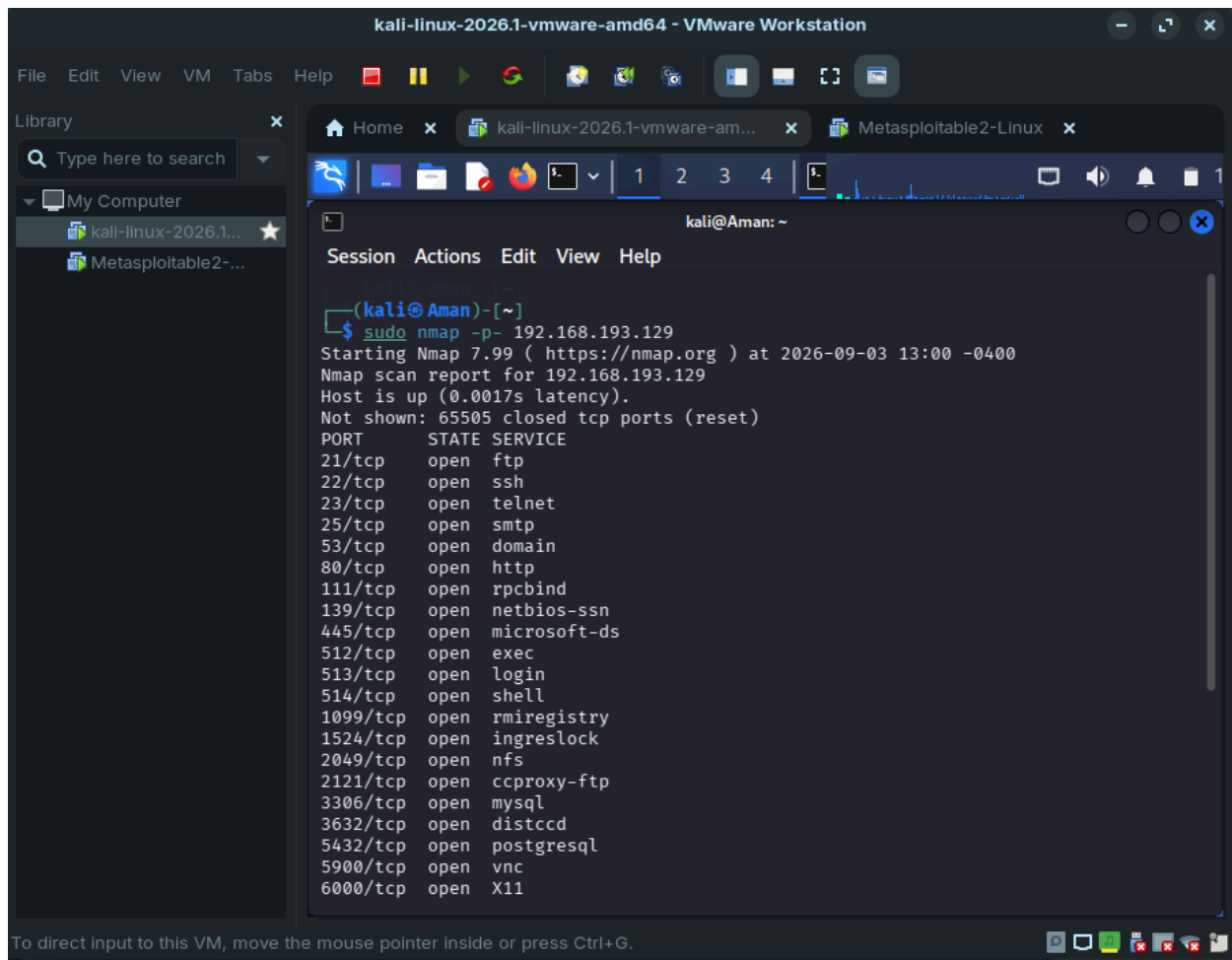


Fig 2N

- I use the command `sudo nmap -A 192.168.193.129` to perform an aggressive scan which combines service version detection, OS fingerprinting and Default NSE script execution as shown in Fig.2L, Fig 2M and Fig. 2N above.

The three extra findings in comparison with step 5 are Anonymous FTP Login, SMB Workgroup/Host Metadata, and HTTP Web Server Methods/Title.

Step 10: Full TCP port scan



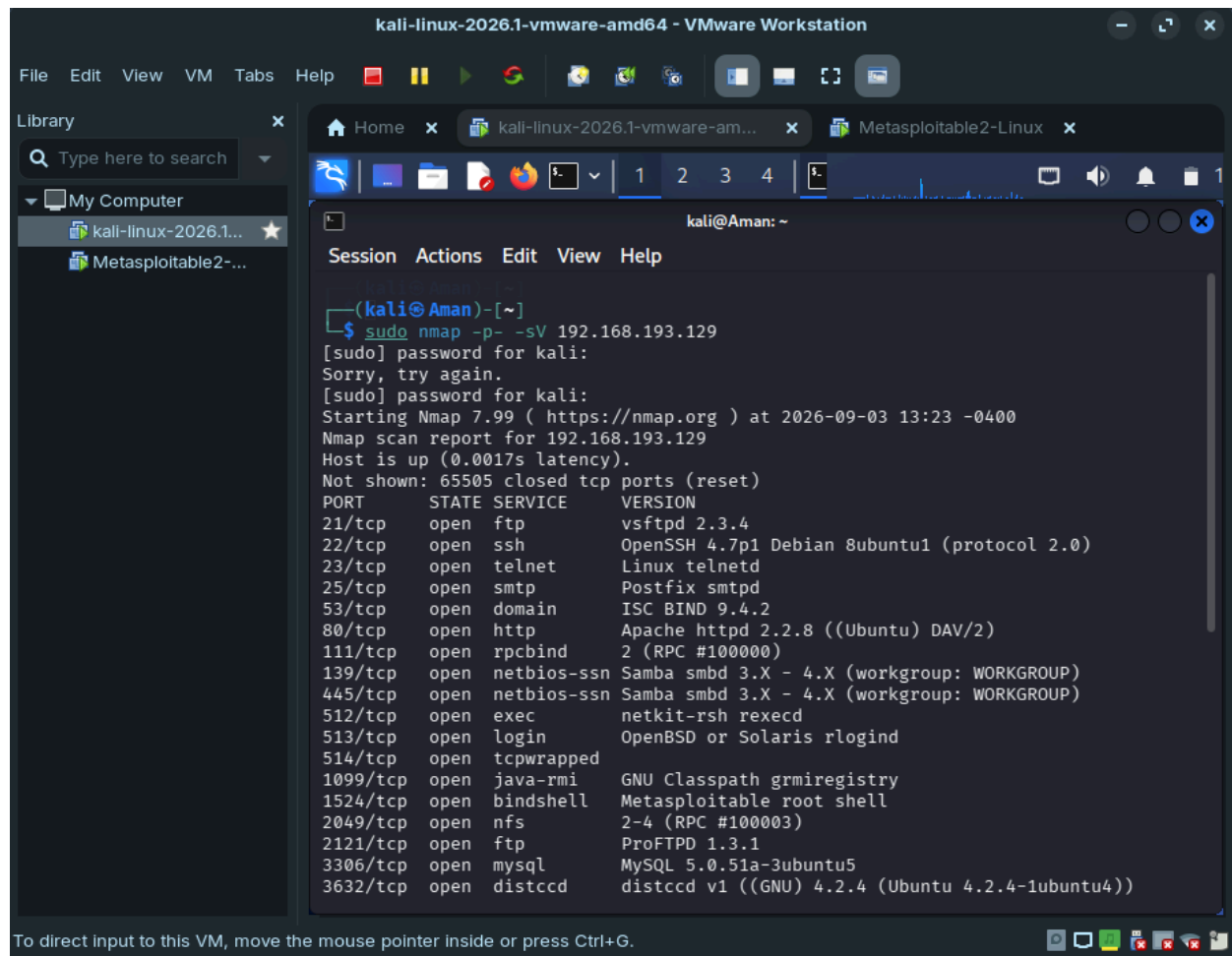
The screenshot shows a Kali Linux virtual machine running in VMware Workstation. A terminal window is open, displaying the results of a full TCP port scan performed using Nmap. The command executed was `sudo nmap -p- 192.168.193.129`. The output indicates that the host is up and lists 30 open ports, each with its corresponding service.

```
(kali@Aman)-[~]
$ sudo nmap -p- 192.168.193.129
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-03 13:00 -0400
Nmap scan report for 192.168.193.129
Host is up (0.0017s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
3632/tcp  open  distccd
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
```

Fig. 20

- Using the command “`sudo nmap -p- <target-ip>`” I perform a Full TCP port scan as shown in Fig. 20 above, the result shows 30 port open making it 7 more additional port open when compared with the other scan that shows 23 port open.

Step 11: Full TCP + versions

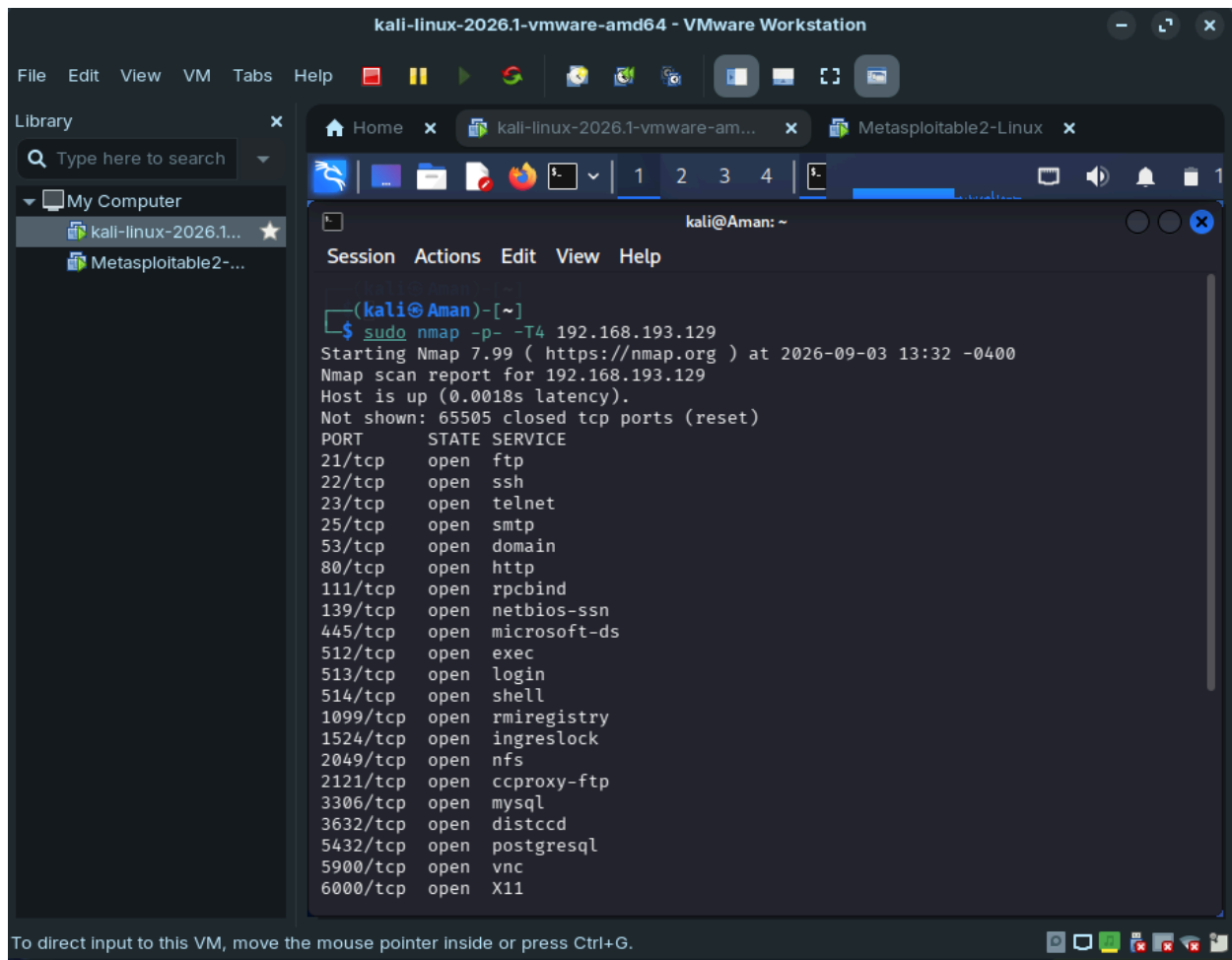


```
kali@Aman: ~  
Session Actions Edit View Help  
  
(kali@Aman)-[~]  
$ sudo nmap -p- -sV 192.168.193.129  
[sudo] password for kali:  
Sorry, try again.  
[sudo] password for kali:  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-03 13:23 -0400  
Nmap scan report for 192.168.193.129  
Host is up (0.0017s latency).  
Not shown: 65505 closed tcp ports (reset)  
PORT      STATE SERVICE      VERSION  
21/tcp    open  ftp          vsftpd 2.3.4  
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)  
23/tcp    open  telnet       Linux telnetd  
25/tcp    open  smtp         Postfix smtpd  
53/tcp    open  domain       ISC BIND 9.4.2  
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)  
111/tcp   open  rpcbind      2 (RPC #100000)  
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)  
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)  
512/tcp   open  exec         netkit-rsh rshcd  
513/tcp   open  login        OpenBSD or Solaris rlogind  
514/tcp   open  tcpwrapped  
1099/tcp  open  java-rmi     GNU Classpath grmiregistry  
1524/tcp  open  bindshell    Metasploitable root shell  
2049/tcp  open  nfs          2-4 (RPC #100003)  
2121/tcp  open  ftp          ProFTPD 1.3.1  
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5  
3632/tcp  open  distccd      distccd v1 ((GNU) 4.2.4 (Ubuntu 4.2.4-1ubuntu4))
```

Fig. 2P

- I used the command “sudo nmap -p- -sV” to get Full TCP with versions as shown in Fig. 2P above.

Step 12: Timing template



The screenshot shows a Kali Linux virtual machine running in VMware Workstation. The terminal window displays the execution of the command `sudo nmap -p- -T4 192.168.193.129`. The output indicates that the scan was completed successfully, showing a list of open ports and their corresponding services. The scan report includes the following information:

```
(kali@Aman)-[~]
$ sudo nmap -p- -T4 192.168.193.129
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-03 13:32 -0400
Nmap scan report for 192.168.193.129
Host is up (0.0018s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
3632/tcp  open  distccd
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
```

Fig. 2Q

- Using the command “`sudo nmap -p- -T4 192.168.193.129`” to get the scan faster as shown in Fig. 2Q and Fig. 2R above. It is 2.17 seconds faster than the normal scan.

Step 13: Selected ports

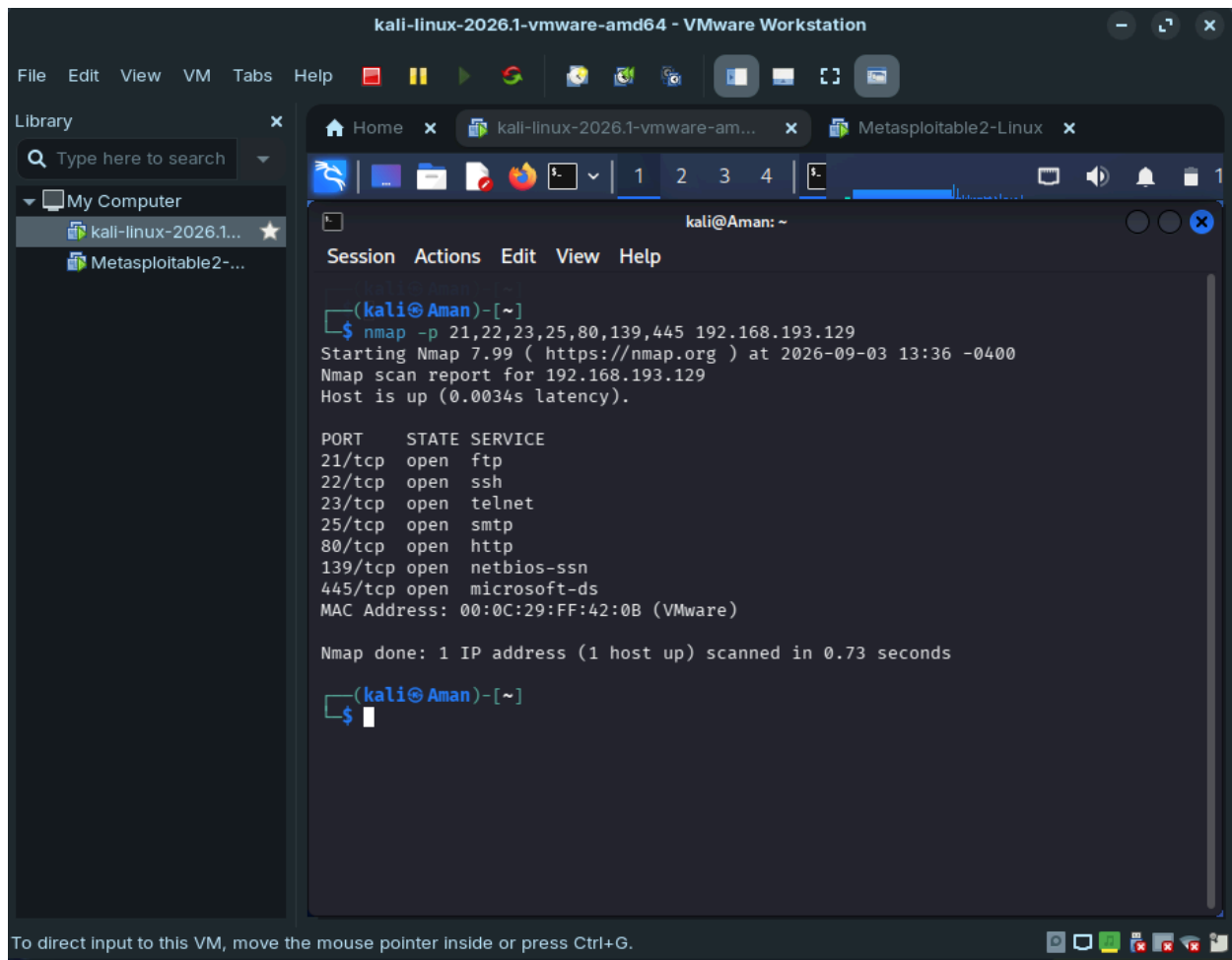


Fig. 2T

- Using the command `nmap -p 21,22,23,25,80,139,445 192.168.193.129` to scan selected ports (21,22,23,25,80,139,445) as shown in Fig. 2T above.

The selected scan is useful For:

- Speed and efficiency during retesting targeted service auditing / Vulnerability verification.
- Minimizing network and log noise.
- Auditing large networks or remote targets over slow or low-bandwidth links.

Step 14: Port range

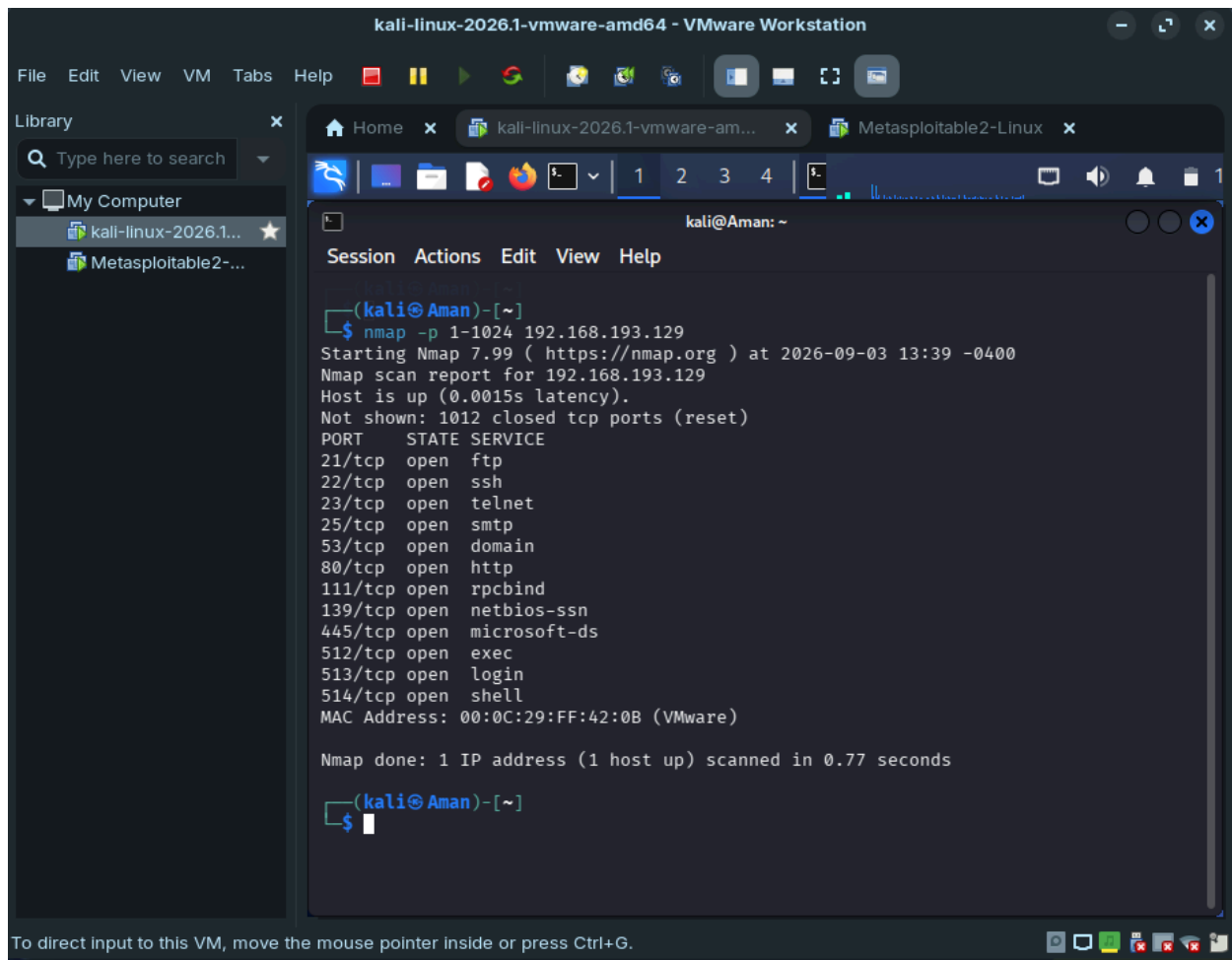
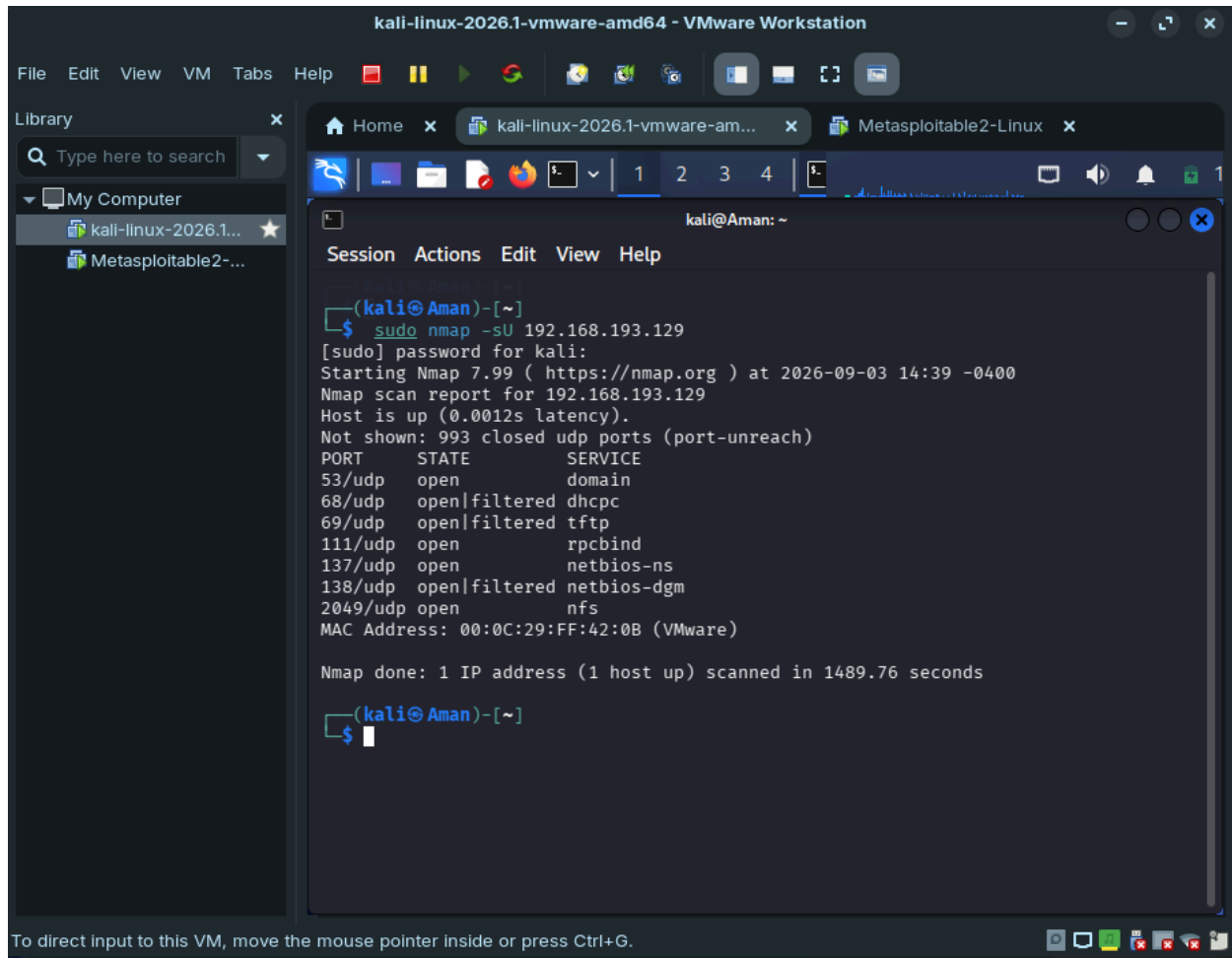


Fig. 2U

- I used the command `nmap -p 1-1024 192.168.129` to scan a range from 1-1024 as shown in Fig. 2U above. This shows 12 ports open compared with the 7 open ports that were shown on step 13.

Step 15: UDP scan



The screenshot shows a Kali Linux virtual machine running in VMware Workstation. The terminal window displays the execution of an Nmap UDP scan. The command used is `sudo nmap -sU 192.168.192.129`. The output shows that the host is up and lists several open UDP ports with their corresponding services.

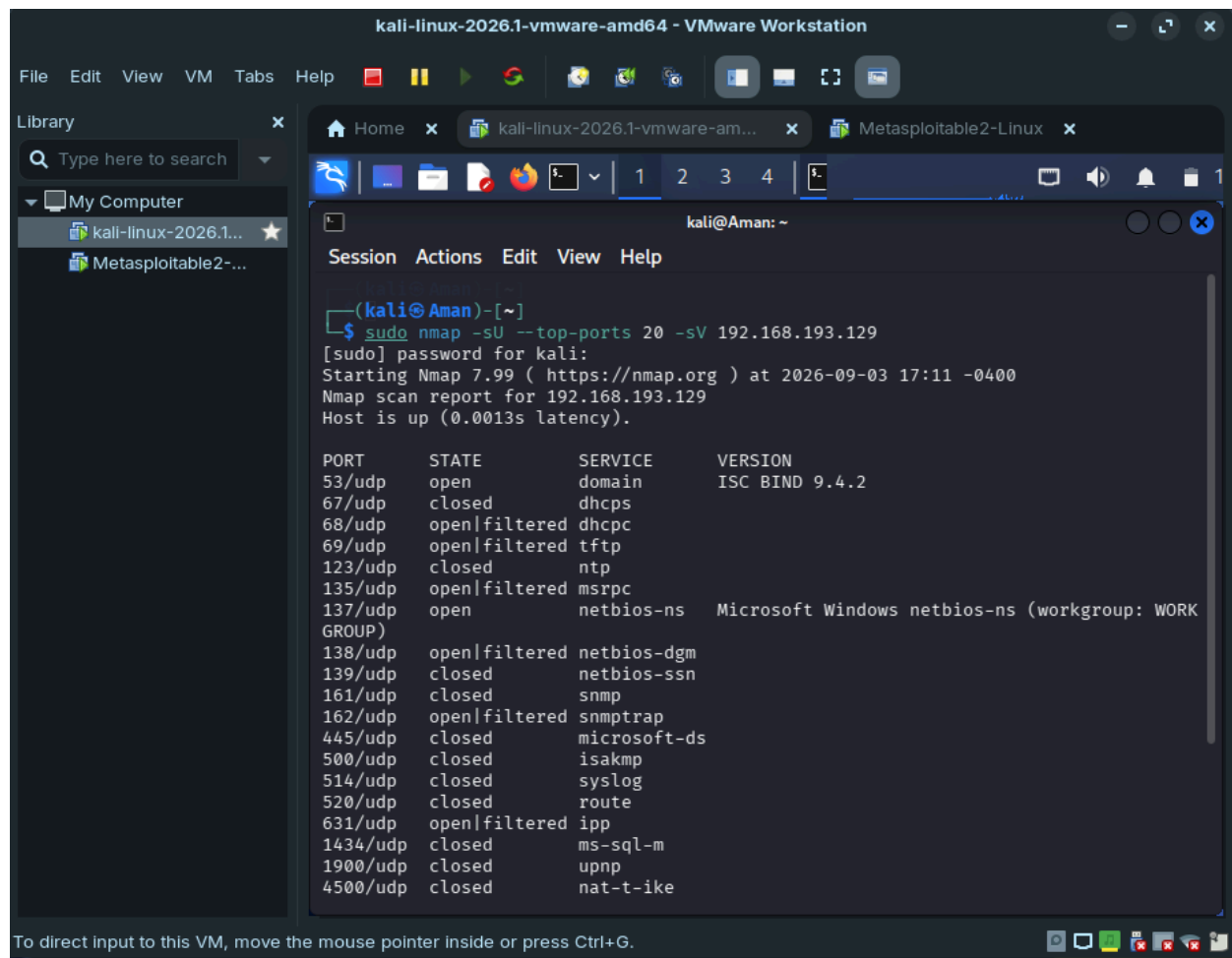
```
(kali@Aman)-[~]
$ sudo nmap -sU 192.168.192.129
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-03 14:39 -0400
Nmap scan report for 192.168.192.129
Host is up (0.0012s latency).
Not shown: 993 closed udp ports (port-unreach)
PORT      STATE      SERVICE
53/udp    open      domain
68/udp    open|filtered dhcpc
69/udp    open|filtered tftp
111/udp   open      rpcbind
137/udp   open      netbios-ns
138/udp   open|filtered netbios-dgm
2049/udp  open      nfs
MAC Address: 00:0C:29:FF:42:0B (VMware)

Nmap done: 1 IP address (1 host up) scanned in 1489.76 seconds
(kali@Aman)-[~]
$
```

Fig. 2V

- Using the command “`sudo nmap -sU 192.168.192.129`” to run udp scan as shown in Fig. 2V above.

Step 16: Top UDP ports + versions



The screenshot shows a Kali Linux virtual machine running in VMware Workstation. The terminal window displays the output of an Nmap scan command: `sudo nmap -sU --top-ports 20 -sV 192.168.193.129`. The scan report shows the host is up and lists the top 20 UDP ports and their associated services and versions.

```
(kali@Aman)-[~]  
$ sudo nmap -sU --top-ports 20 -sV 192.168.193.129  
[sudo] password for kali:  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-03 17:11 -0400  
Nmap scan report for 192.168.193.129  
Host is up (0.0013s latency).  
  
PORT      STATE      SERVICE      VERSION  
53/udp    open       domain       ISC BIND 9.4.2  
67/udp    closed     dhcpd          
68/udp    open|filtered dhcpd          
69/udp    open|filtered tftp          
123/udp   closed     ntp            
135/udp   open|filtered msrpc         
137/udp   open       netbios-ns   Microsoft Windows netbios-ns (workgroup: WORK  
GROUP)  
138/udp   open|filtered netbios-dgm    
139/udp   closed     netbios-ssn    
161/udp   closed     snmp           
162/udp   open|filtered snmptrap       
445/udp   closed     microsoft-ds   
500/udp   closed     isakmp         
514/udp   closed     syslog         
520/udp   closed     route          
631/udp   open|filtered ipp            
1434/udp  closed     ms-sql-m       
1900/udp  closed     upnp           
4500/udp  closed     nat-t-ike    
```

Fig 2W

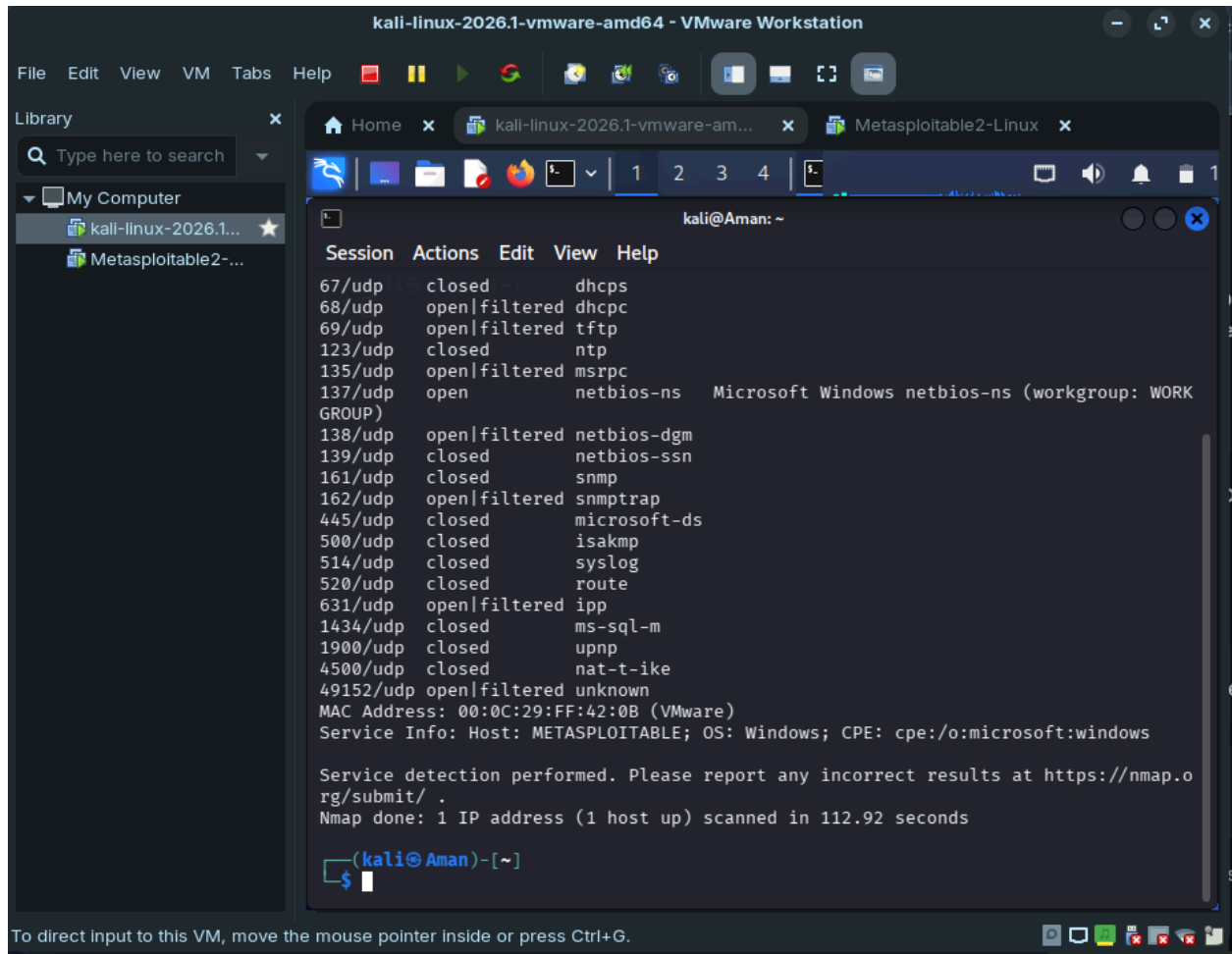
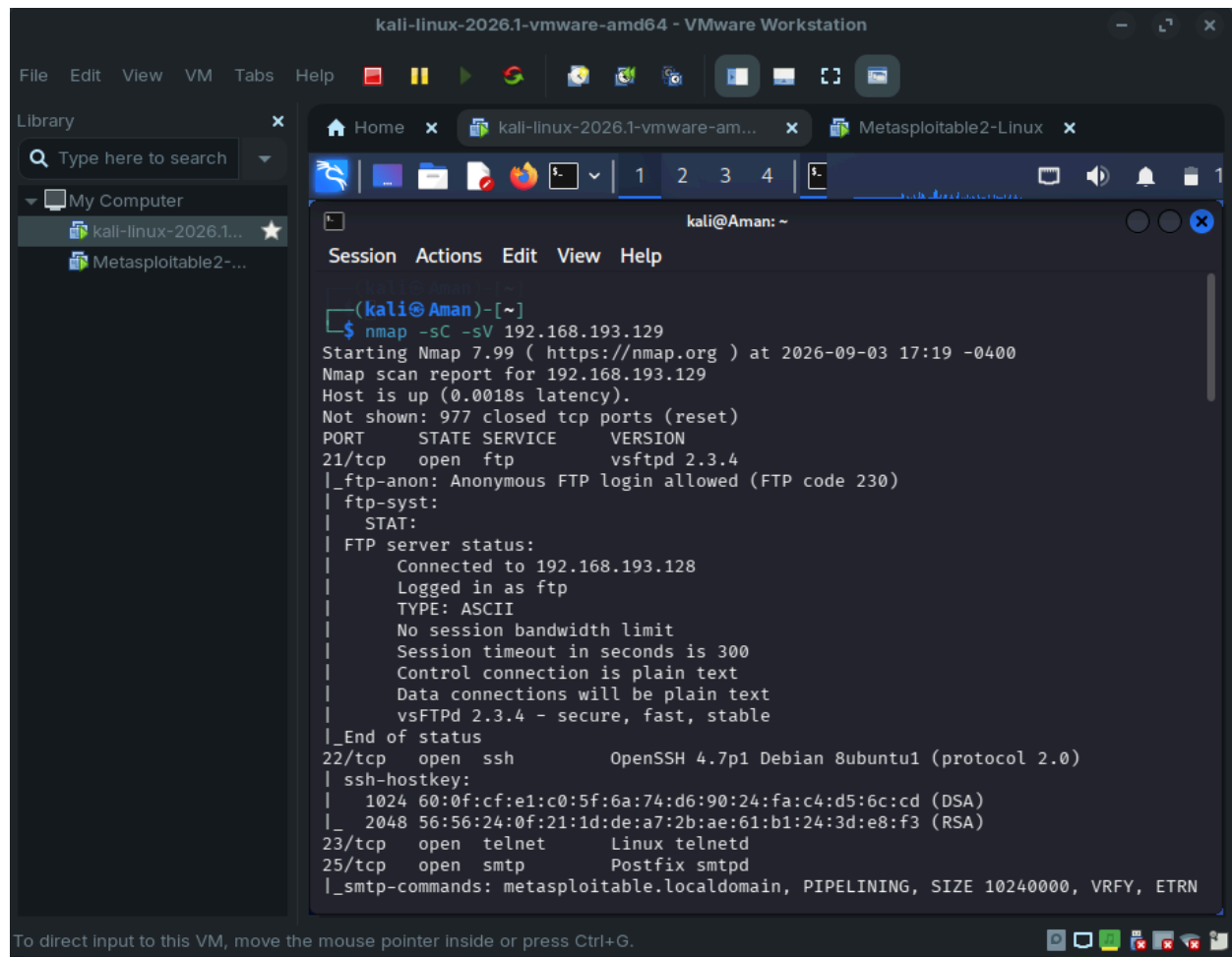


Fig. 2X

- I used the command “sudo nmap -sU --top-ports 20 -sV 192.168.193.129 to scan common UDP sets and identify services as shown in Fig. 2W and Fig. 2X.

Part 4 - Safe Nmap NSE Enumeration

Step 17: Default scripts



```
kali-linux-2026.1-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
kali-linux-2026.1...
Metasploitable2-...

kali@Aman: ~
Session Actions Edit View Help

(kali@Aman)-[~]
$ nmap -sC -sV 192.168.193.129
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-03 17:19 -0400
Nmap scan report for 192.168.193.129
Host is up (0.0018s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_ftp-syst:
|_STAT:
|_FTP server status:
|_   Connected to 192.168.193.128
|_   Logged in as ftp
|_   TYPE: ASCII
|_   No session bandwidth limit
|_   Session timeout in seconds is 300
|_   Control connection is plain text
|_   Data connections will be plain text
|_   vsFTPD 2.3.4 - secure, fast, stable
|_End of status
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ssh-hostkey:
|_   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_   2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
|_smtp_commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN
```

Fig. 3A

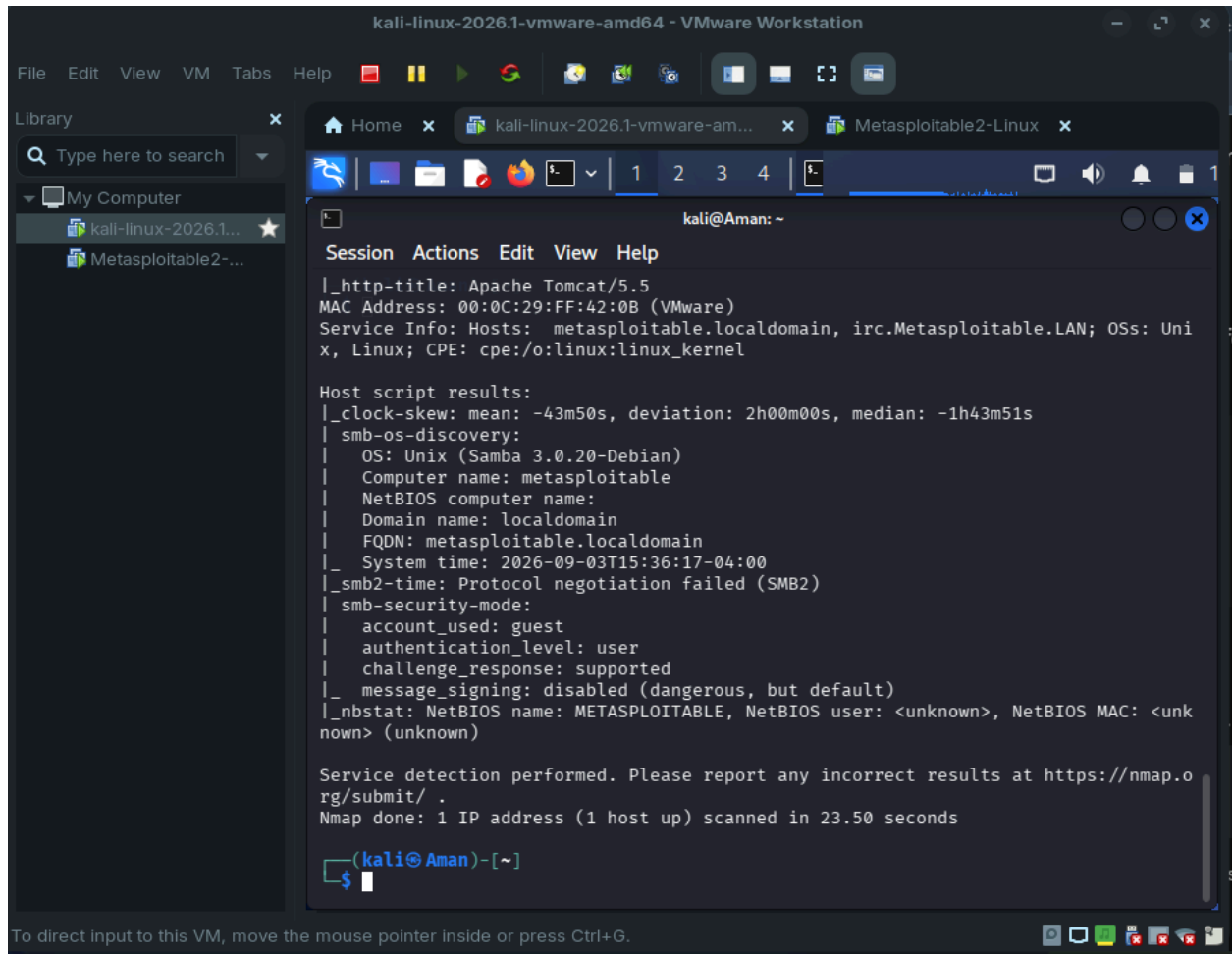


Fig. 3B

I used the command `nmap -sC -sV 192.168.193.129` to collect common service information. As shown in Fig. 3B above.

Step 18: HTTP title

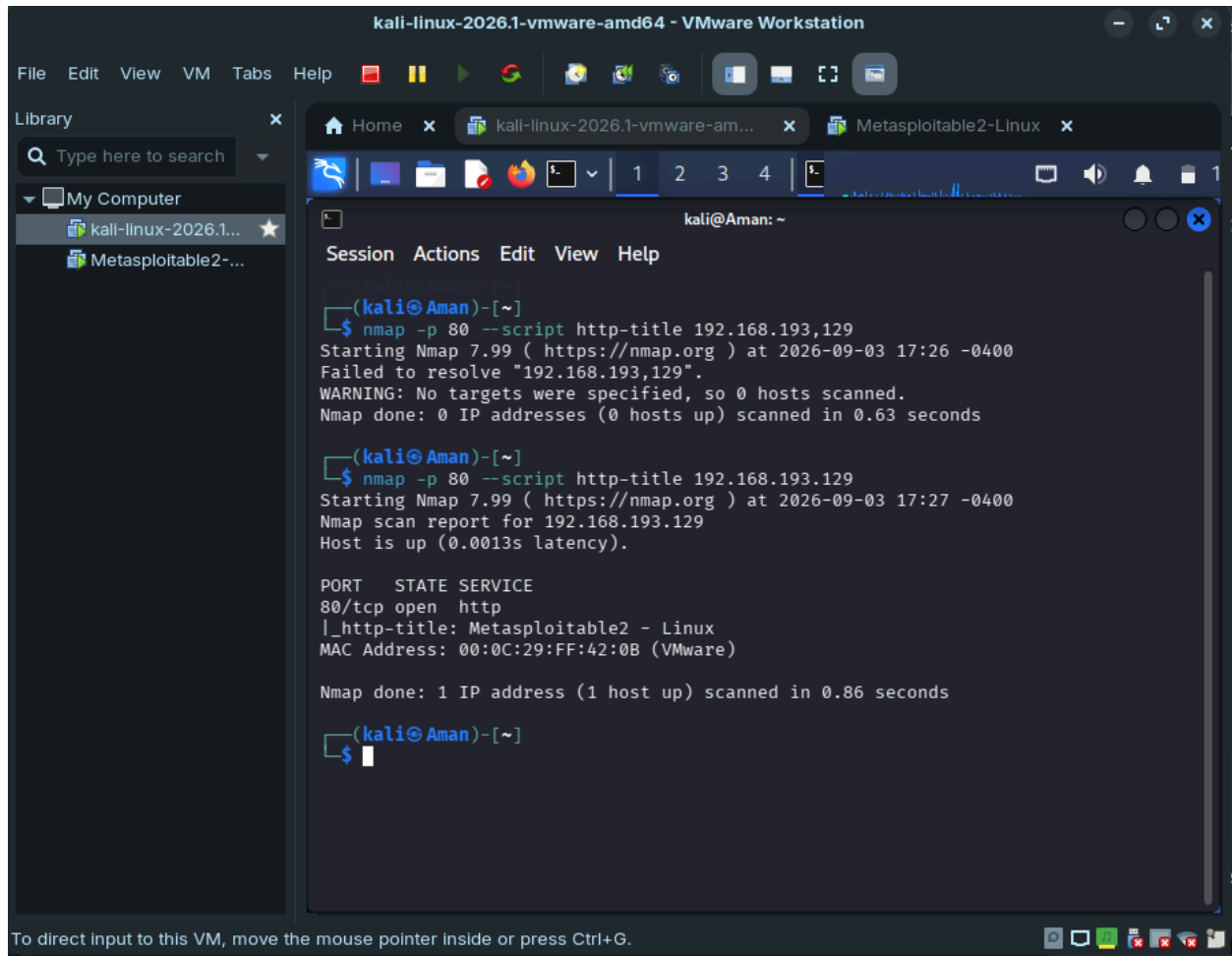
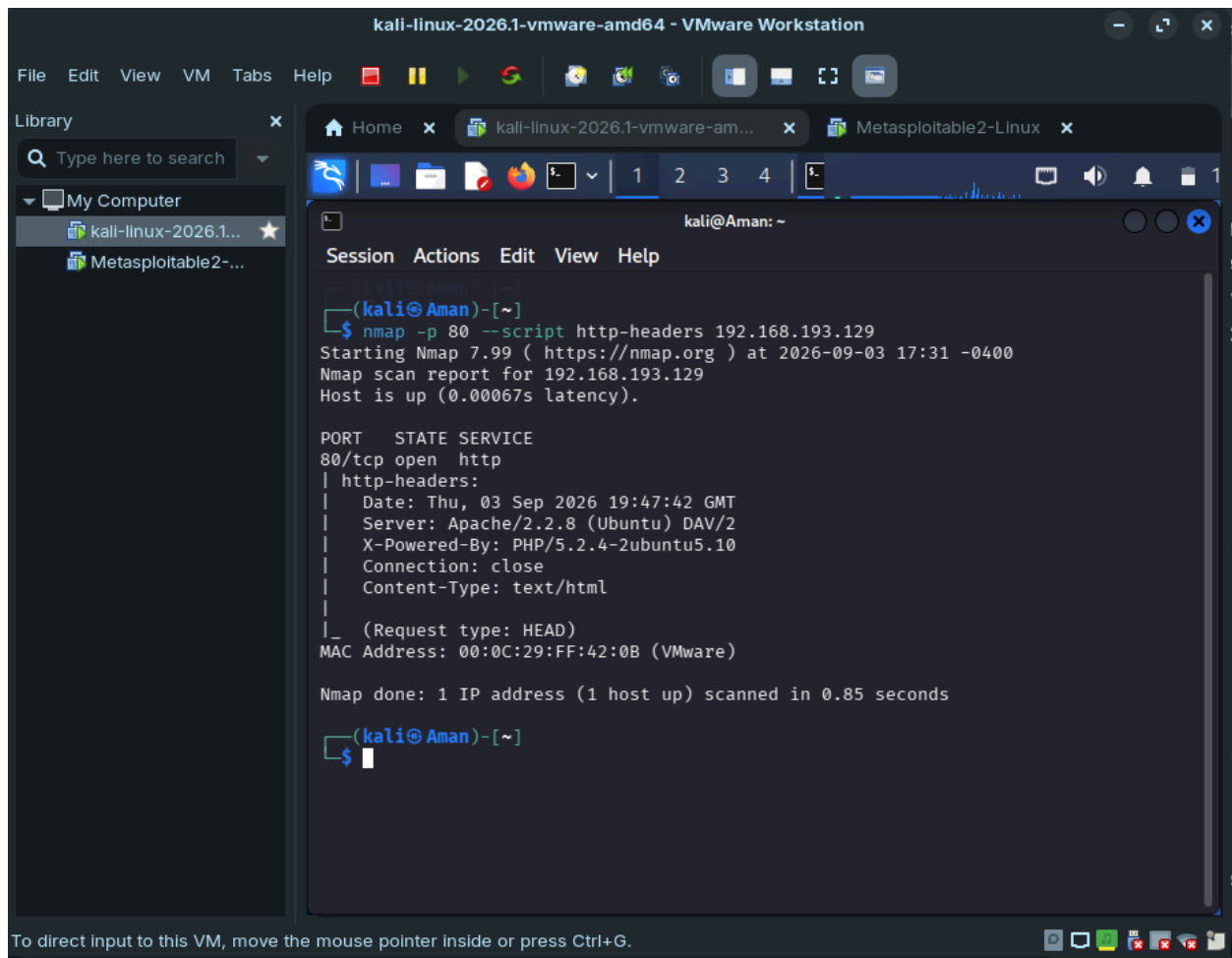


Fig. 3C

- Using the command `nmap -p 80 --script http-title 192.168.193.129` to view the HTTP title: Metasploitable2 - Linux.

Step 19: HTTP headers



The screenshot shows a Kali Linux virtual machine running in VMware Workstation. The terminal window displays the following output:

```
(kali@Aman)-[~]
$ nmap -p 80 --script http-headers 192.168.193.129
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-03 17:31 -0400
Nmap scan report for 192.168.193.129
Host is up (0.00067s latency).

PORT      STATE SERVICE
80/tcp    open  http
| http-headers:
|   Date: Thu, 03 Sep 2026 19:47:42 GMT
|   Server: Apache/2.2.8 (Ubuntu) DAV/2
|   X-Powered-By: PHP/5.2.4-2ubuntu5.10
|   Connection: close
|   Content-Type: text/html
|
|_ (Request type: HEAD)
MAC Address: 00:0C:29:FF:42:0B (VMware)

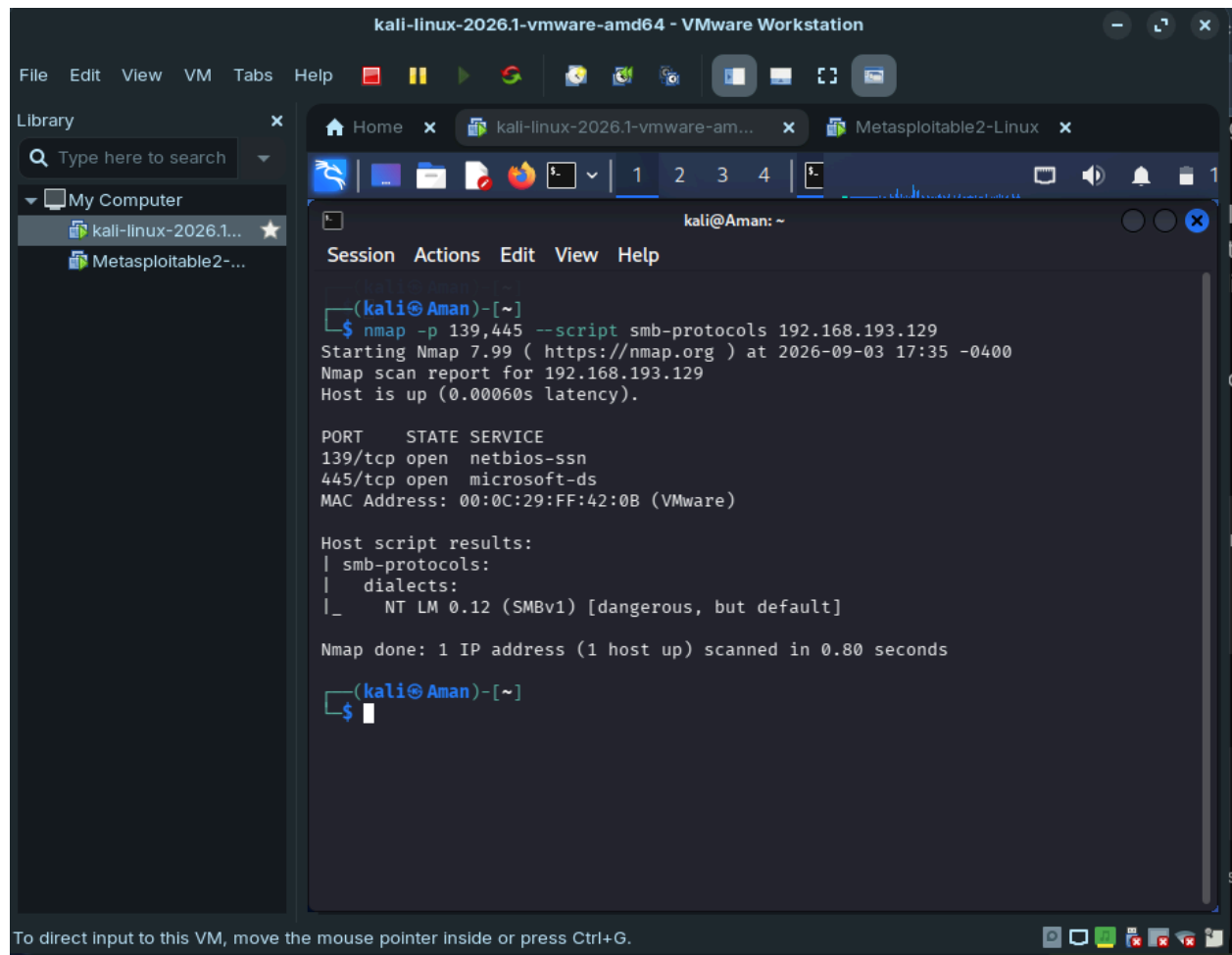
Nmap done: 1 IP address (1 host up) scanned in 0.85 seconds

(kali@Aman)-[~]
$
```

Fig. 3D

- Using the command “`nmap -p 80 --script http-headers 192.168.193.129`” to identify HTTP headers as shown in Fig. 3D above.

Step 20: SMB protocols



The screenshot shows a VMware Workstation window titled "kali-linux-2026.1-vmware-amd64 - VMware Workstation". Inside, a terminal window titled "kali@Aman: ~" is open. The terminal displays the following output:

```
(kali@Aman)-[~]
$ nmap -p 139,445 --script smb-protocols 192.168.193.129
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-03 17:35 -0400
Nmap scan report for 192.168.193.129
Host is up (0.00060s latency).

PORT      STATE SERVICE
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
MAC Address: 00:0C:29:FF:42:0B (VMware)

Host script results:
| smb-protocols:
|   dialects:
|_    NT LM 0.12 (SMBv1) [dangerous, but default]

Nmap done: 1 IP address (1 host up) scanned in 0.80 seconds

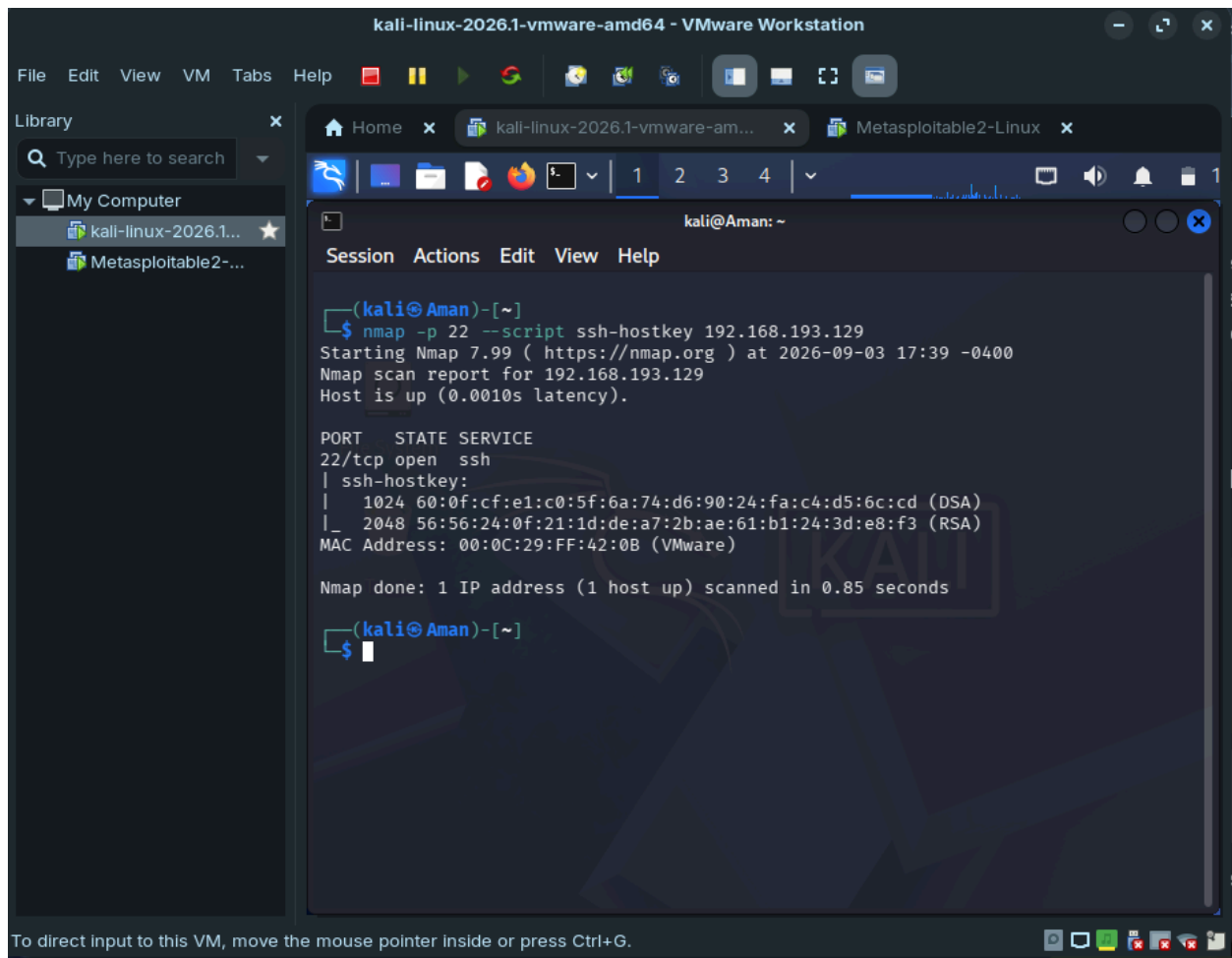
(kali@Aman)-[~]
$
```

At the bottom of the VMware window, a message reads: "To direct input to this VM, move the mouse pointer inside or press Ctrl+G."

Fig. 3E

- Using the command `nmap -p 139,445 --script smb-protocols 192.168.193.129` to inspect supported SMB protocols as shown in Fig. 3E

Step 21: SSH host keys



```
kali@Aman: ~  
$ nmap -p 22 --script ssh-hostkey 192.168.193.129  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-03 17:39 -0400  
Nmap scan report for 192.168.193.129  
Host is up (0.0010s latency).  
  
PORT      STATE SERVICE  
22/tcp    open  ssh  
| ssh-hostkey:  
|_ 1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)  
|_ 2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)  
MAC Address: 00:0C:29:FF:42:0B (VMware)  
  
Nmap done: 1 IP address (1 host up) scanned in 0.85 seconds  
  
kali@Aman: ~  
$
```

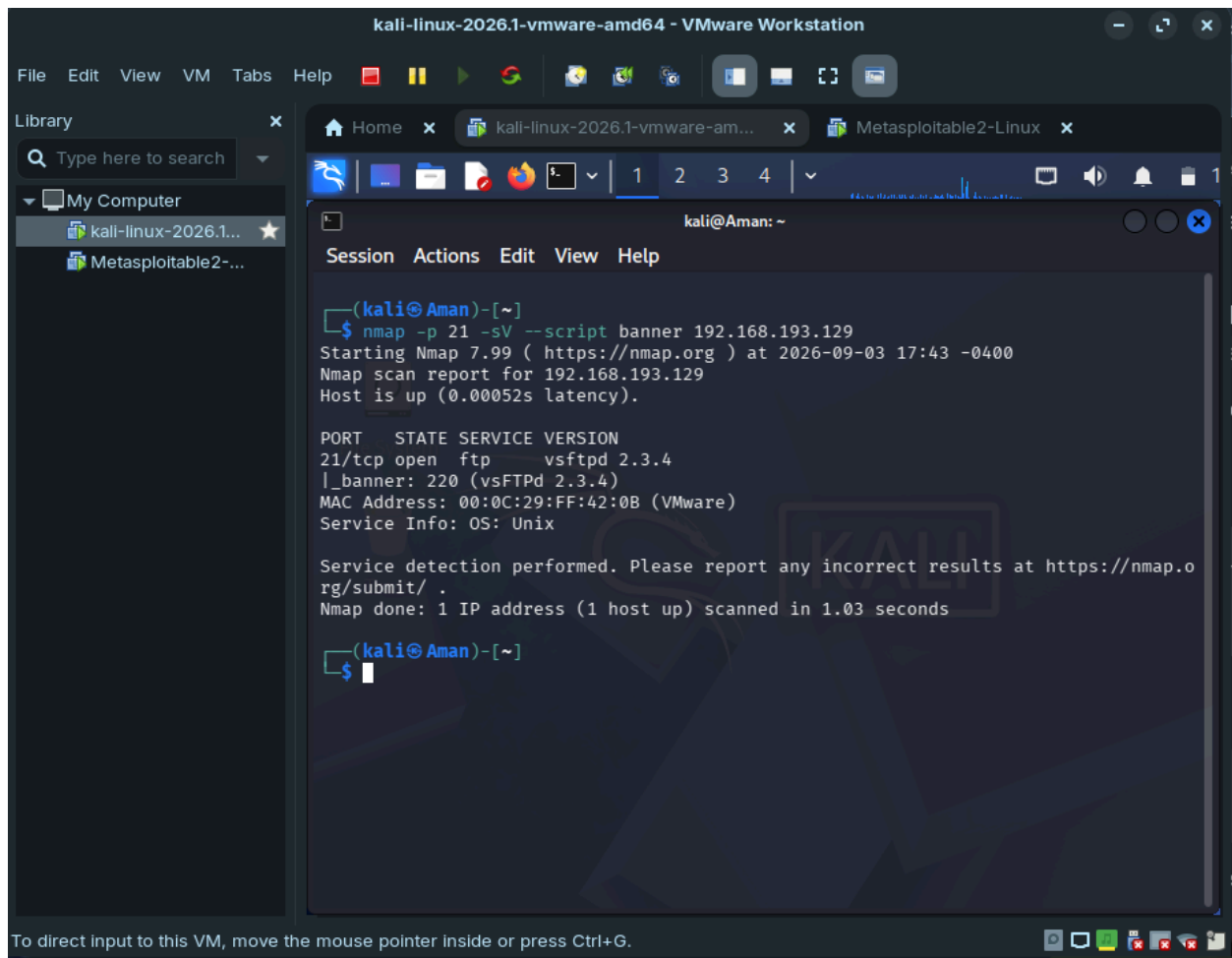
Fig. 3F

- Using the command “nmap -p 22 --script ssh-hostkey 192.168.193.129” SSH servers publish host keys as shown in Fig. 3F above.

Algorithm Names Identified:

- DSA (1024-bit Digital Signature Algorithm)
- RSA (2048-bit Rivest–Shamir–Adleman)

Step 22: FTP banner



```
kali@Aman: ~  
$ nmap -p 21 -sV --script banner 192.168.193.129  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-03 17:43 -0400  
Nmap scan report for 192.168.193.129  
Host is up (0.00052s latency).  
  
PORT      STATE SERVICE VERSION  
21/tcp    open  ftp      vsftpd 2.3.4  
|_banner: 220 (vsFTPd 2.3.4)  
MAC Address: 00:0C:29:FF:42:0B (VMware)  
Service Info: OS: Unix  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 1.03 seconds  
  
kali@Aman: ~  
$
```

Fig. 3G

- I used the command “nmap -p 21 -sV --script banner 192.168.193.129” to view the service banner, as shown in Fig. 3G above.

When filling out your lab report, compare the two output fields:

- **Raw Banner Output (_banner):** 220 (vsFTPd 2.3.4)
- **Parsed Version Output (VERSION):** vsftpd 2.3.4

Part 5 - Manual HTTP Confirmation

Step 23: Request headers with curl

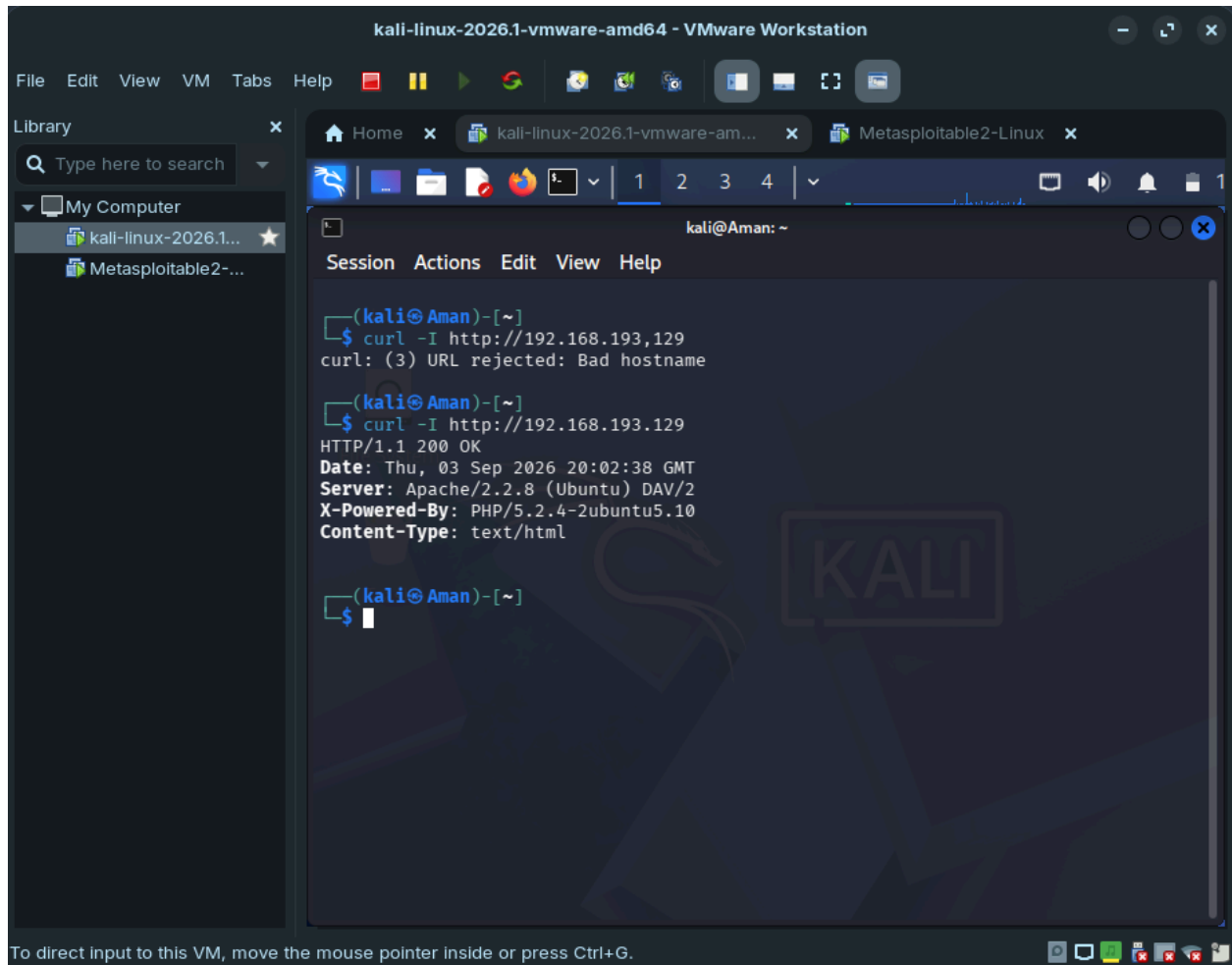


Fig. 4A

- I used the command `curl -I http 192.168.193.129` to find http status and headers as shown in Fig. 4A above.

HTTP Status Code: 200 OK

Server Header: Apache/2.2.8 (Ubuntu) DAV/2

Part 6 - WhatWeb Fingerprinting

Step 24: Basic fingerprint

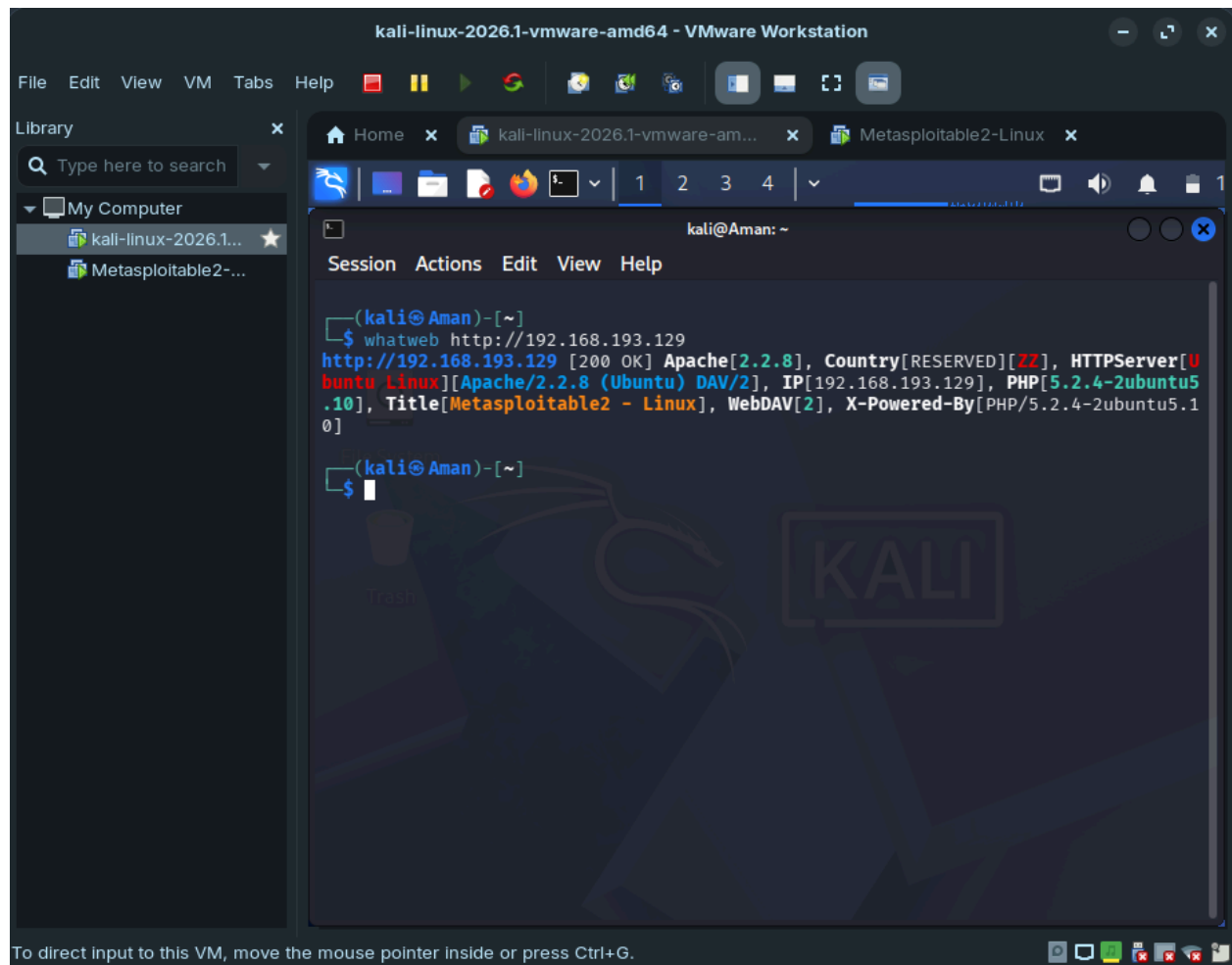


Fig. 5A

- I used the command “whatweb http://<target-ip>” to view the compact technology summary as shown in Fig. 5A and find these technologies used :

Web Server: Apache 2.2.8

Operating System / Distribution: Ubuntu Linux

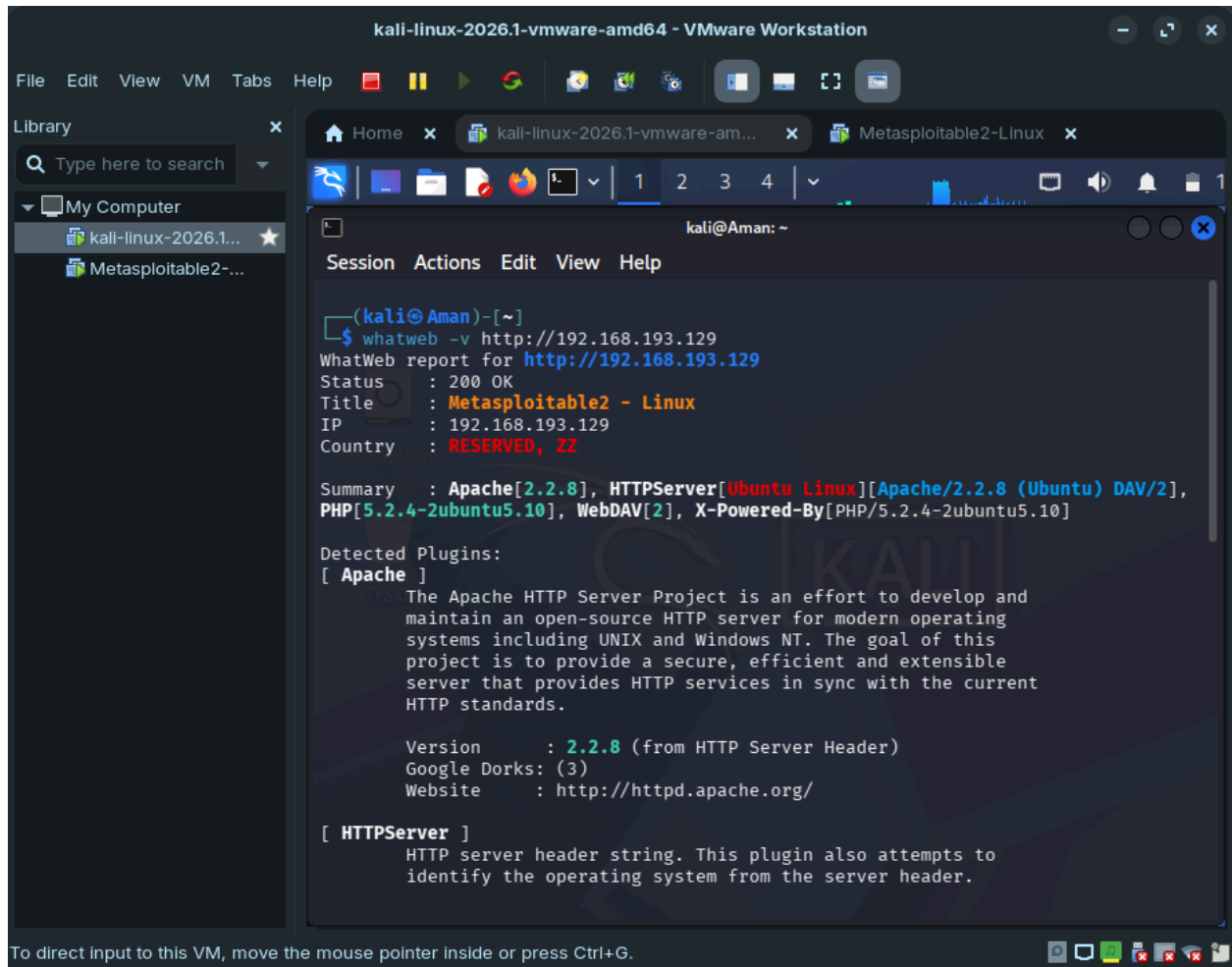
Server-Side Scripting Language: PHP 5.2.4 (Build: 5.2.4-2ubuntu5.10)

Web Server Extensions: WebDAV (DAV/2)

HTML Page Title: Metasploitable2 - Linux

HTTP Response Code: 200 OK

Step 25: Verbose output



```
kali-linux-2026.1-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
kali-linux-2026.1...
Metasploitable2-...

kali@Aman: ~
Session Actions Edit View Help

(kali@Aman)-[~]
$ whatweb -v http://192.168.193.129
WhatWeb report for http://192.168.193.129
Status : 200 OK
Title : Metasploitable2 - Linux
IP : 192.168.193.129
Country : RESERVED, ZZ

Summary : Apache[2.2.8], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2],
PHP[5.2.4-2ubuntu5.10], WebDAV[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]

Detected Plugins:
[ Apache ]
The Apache HTTP Server Project is an effort to develop and
maintain an open-source HTTP server for modern operating
systems including UNIX and Windows NT. The goal of this
project is to provide a secure, efficient and extensible
server that provides HTTP services in sync with the current
HTTP standards.

Version : 2.2.8 (from HTTP Server Header)
Google Dorks: (3)
Website : http://httpd.apache.org/

[ HTTPServer ]
HTTP server header string. This plugin also attempts to
identify the operating system from the server header.
```

Fig. 5B

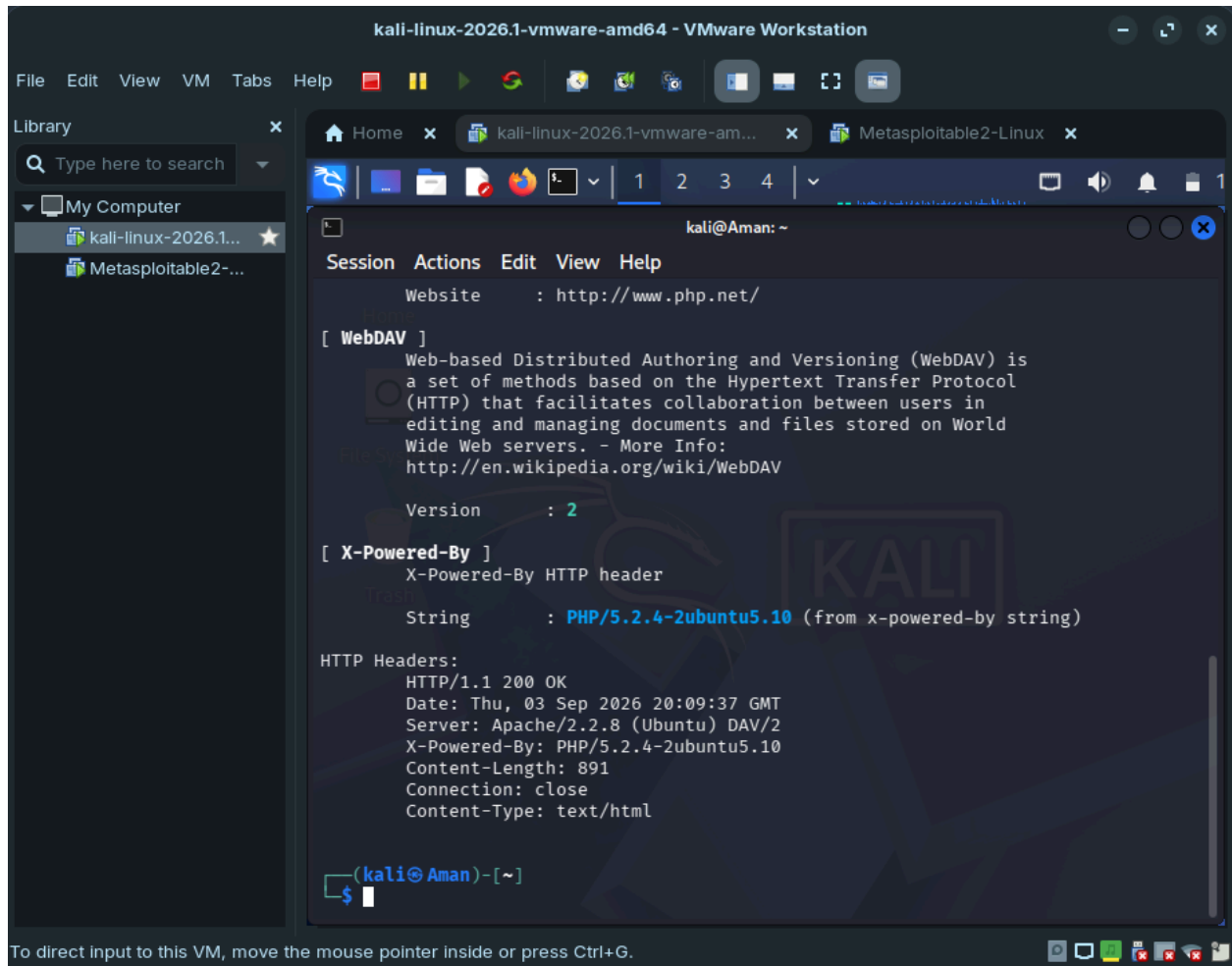


Fig. 5C

- I used the command `whatweb -v http://192.168.193.129` to produce a more detailed result. As shown in Fig. 5B and Fig. 5C above.

The verbose switch (-v) expands the compact fingerprint into individual plugin assessments, mentioning how each technology was recognized.

Step 26: Aggression level 1

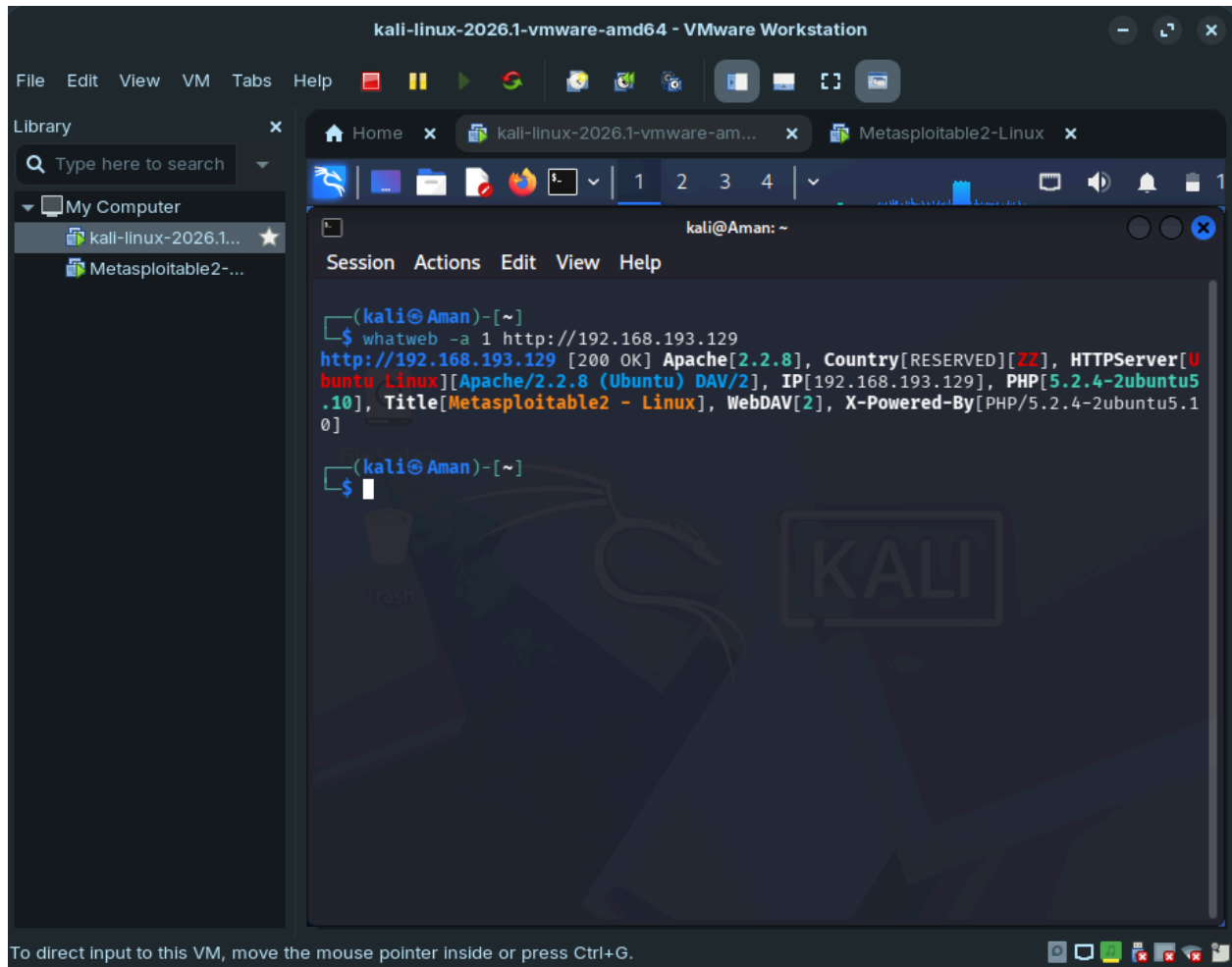


Fig 5D

- I used the command “whatweb -a 1 <http://192.168.193.129>” to make the fingerprint aggression level 1 as shown in Fig. 5D above.

Discovered Stack:

- Web Server: Apache 2.2.8
- Language: PHP 5.2.4-2ubuntu5.10
- Modules: WebDAV 2
- Operating System: Ubuntu Linux

Step 27: Aggression level 3

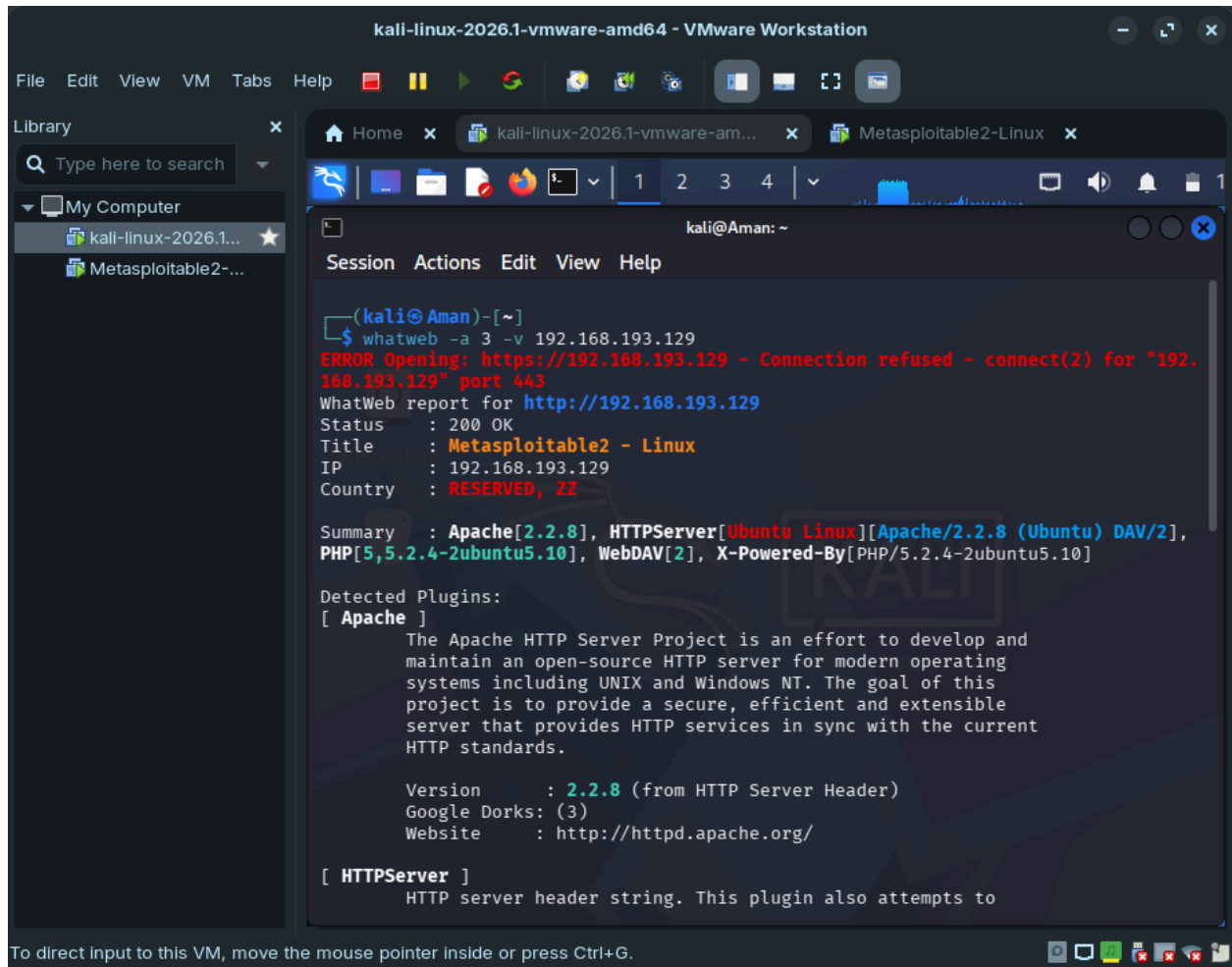


Fig. 5E

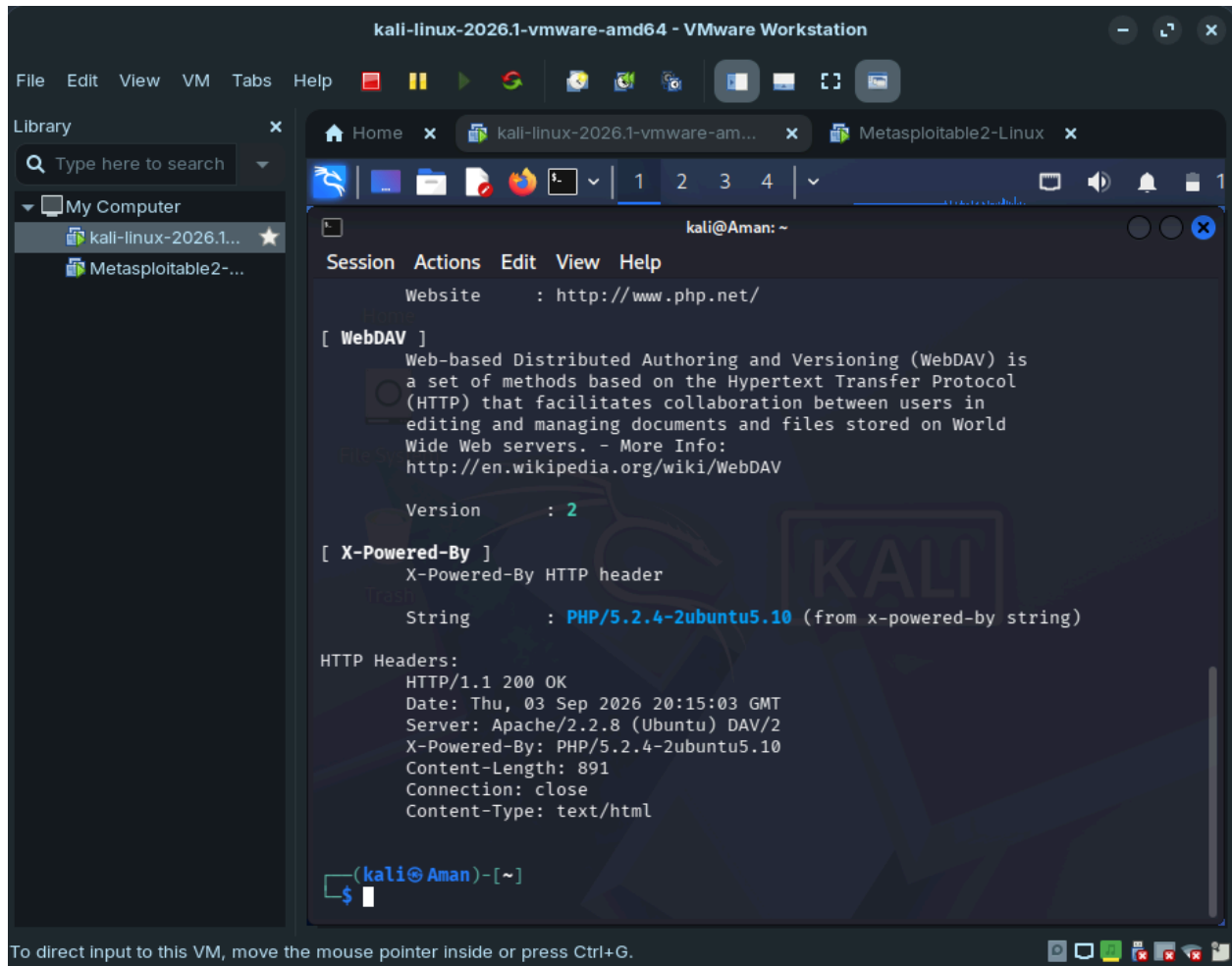


Fig. 5F

I used the command “whatweb -a 3 -v <http://192.168.129>” to increase the aggression level to 3 and view more detailed identification as shown in Fig. 5E and Fig. 5F above.

Aggressive Mode (Level 3) Behavior:

- Level 3 triggers additional recursive link extraction and deeper plugin heuristics compared to passive Level 1 scans.
- PHP Detection: Matched both the explicit version string (5.2.4-2ubuntu5.10) and extracted the major branch release (PHP 5).

Step 28: Aggression level 4

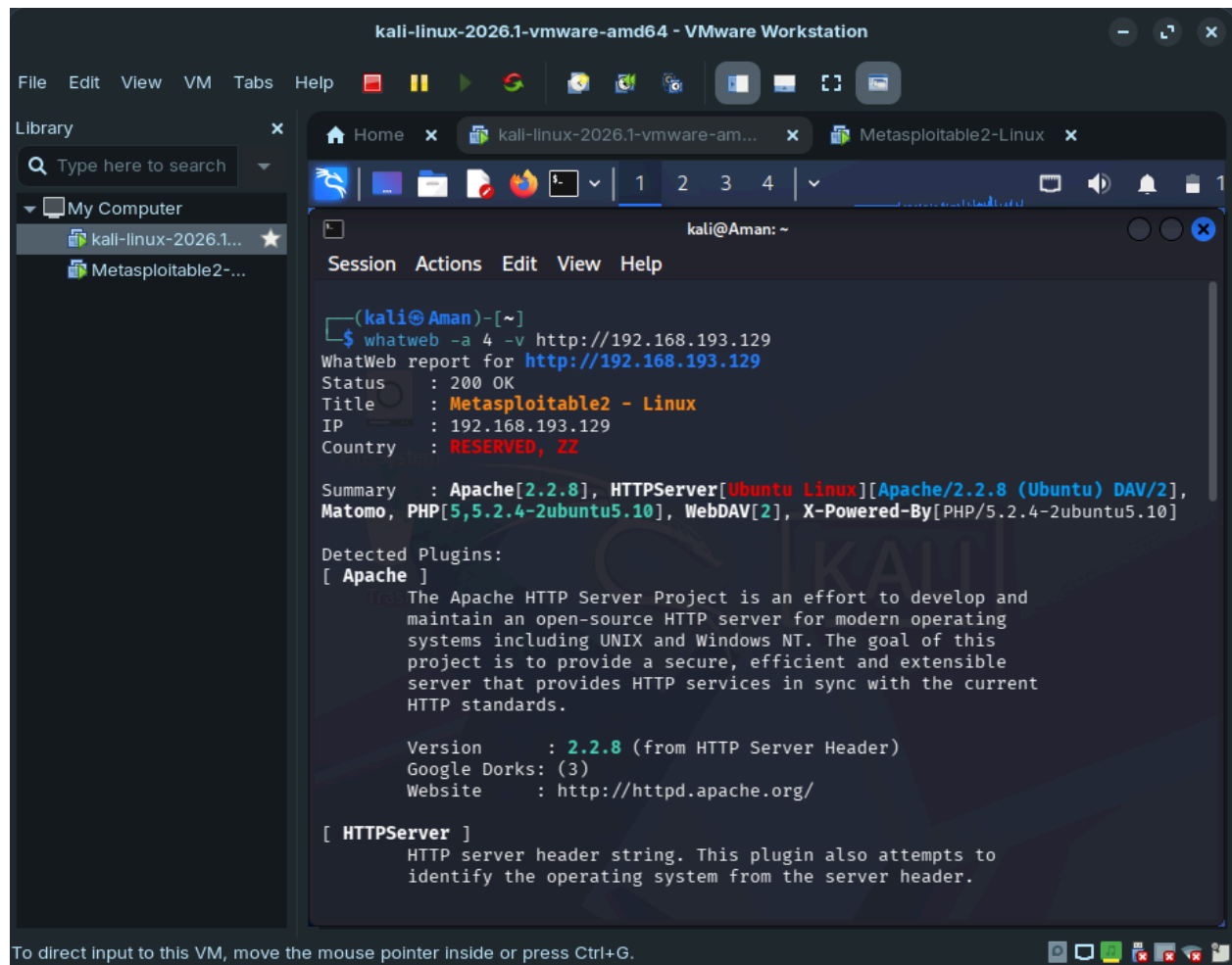


Fig. 5G

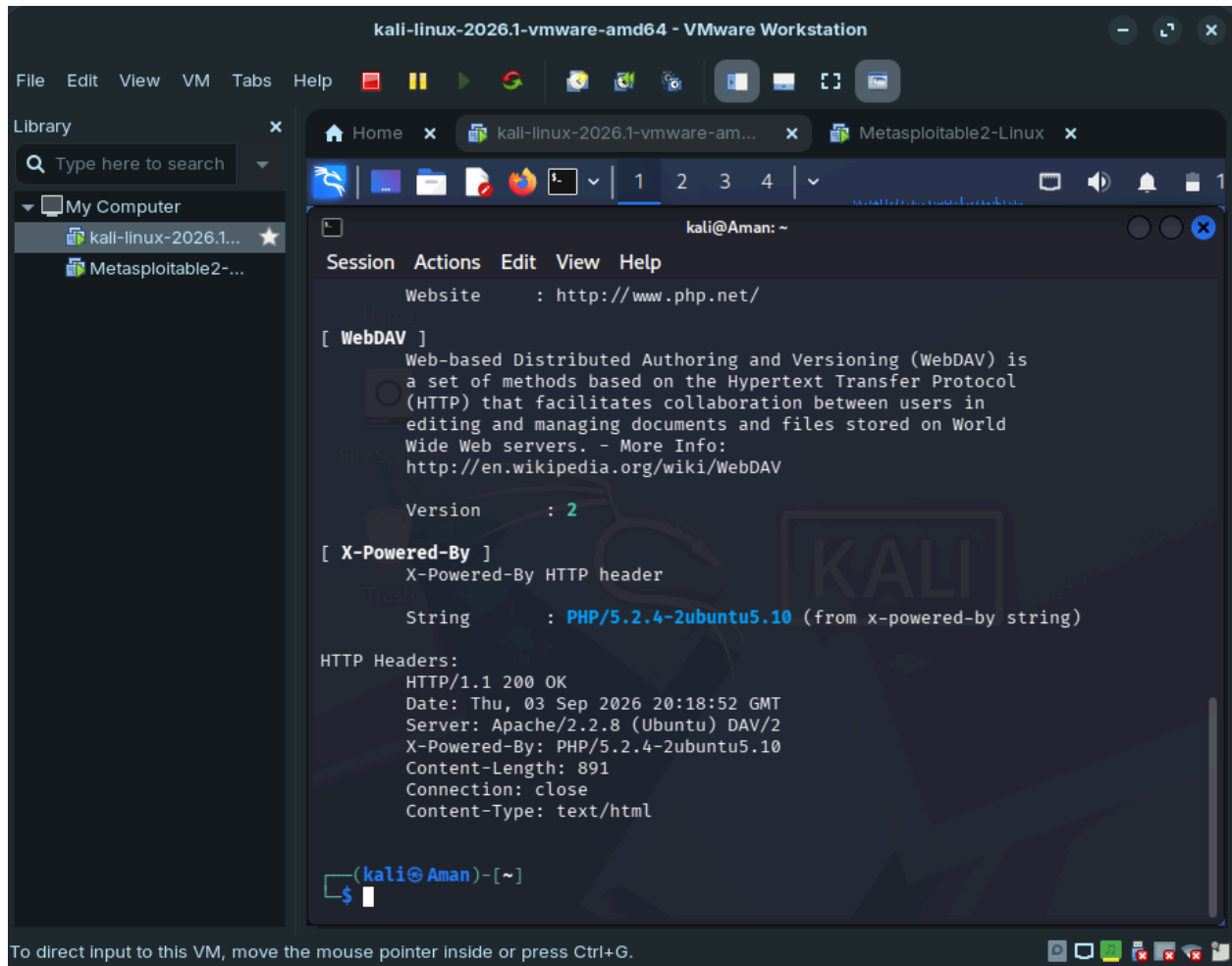


Fig 5H

Using the command `whatweb -a 4 -v http://192.168.193.129` and raise the aggression level to 4 which takes longer time and generates more requests and details.

Step 29: Follow redirects

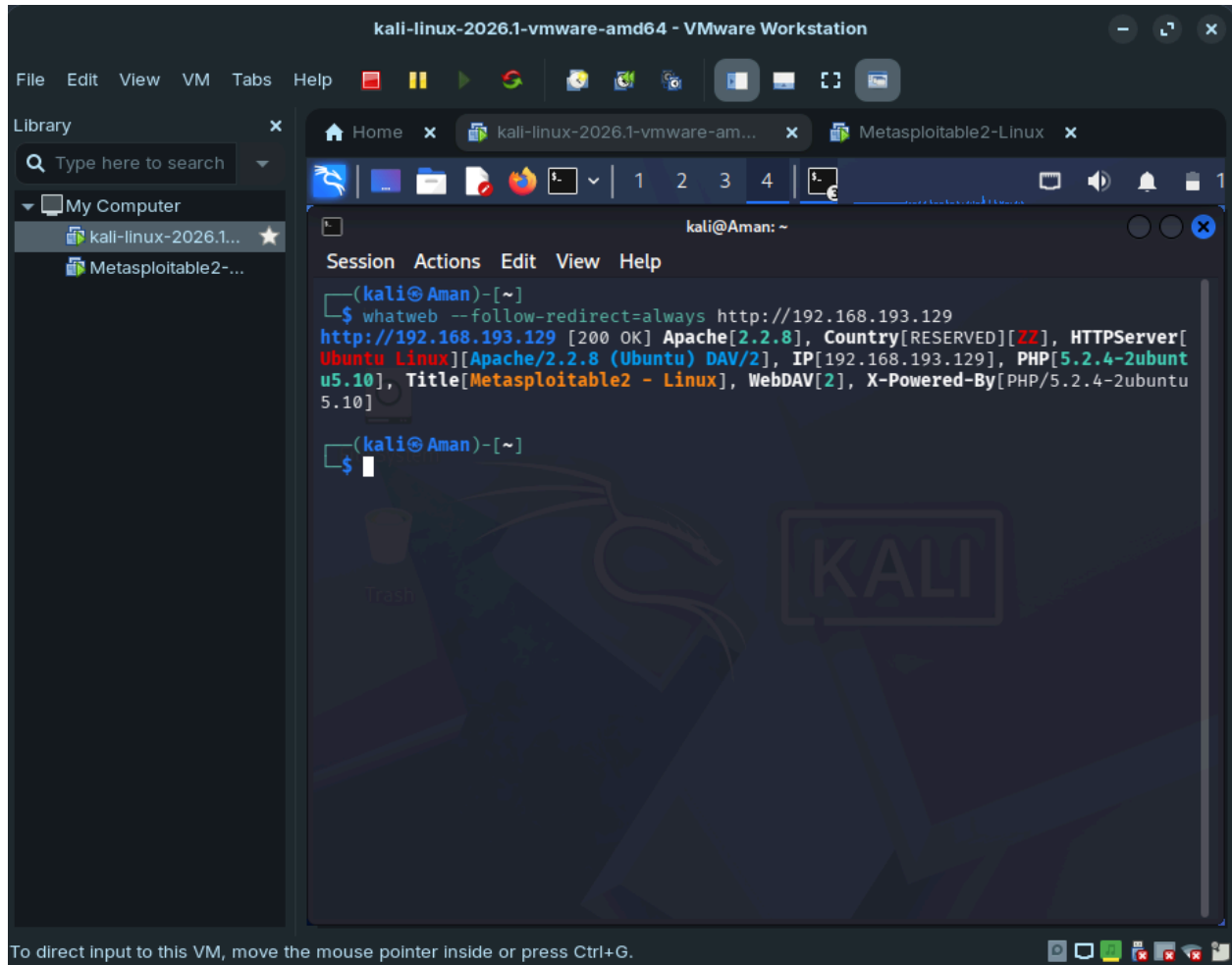


Fig. 5I

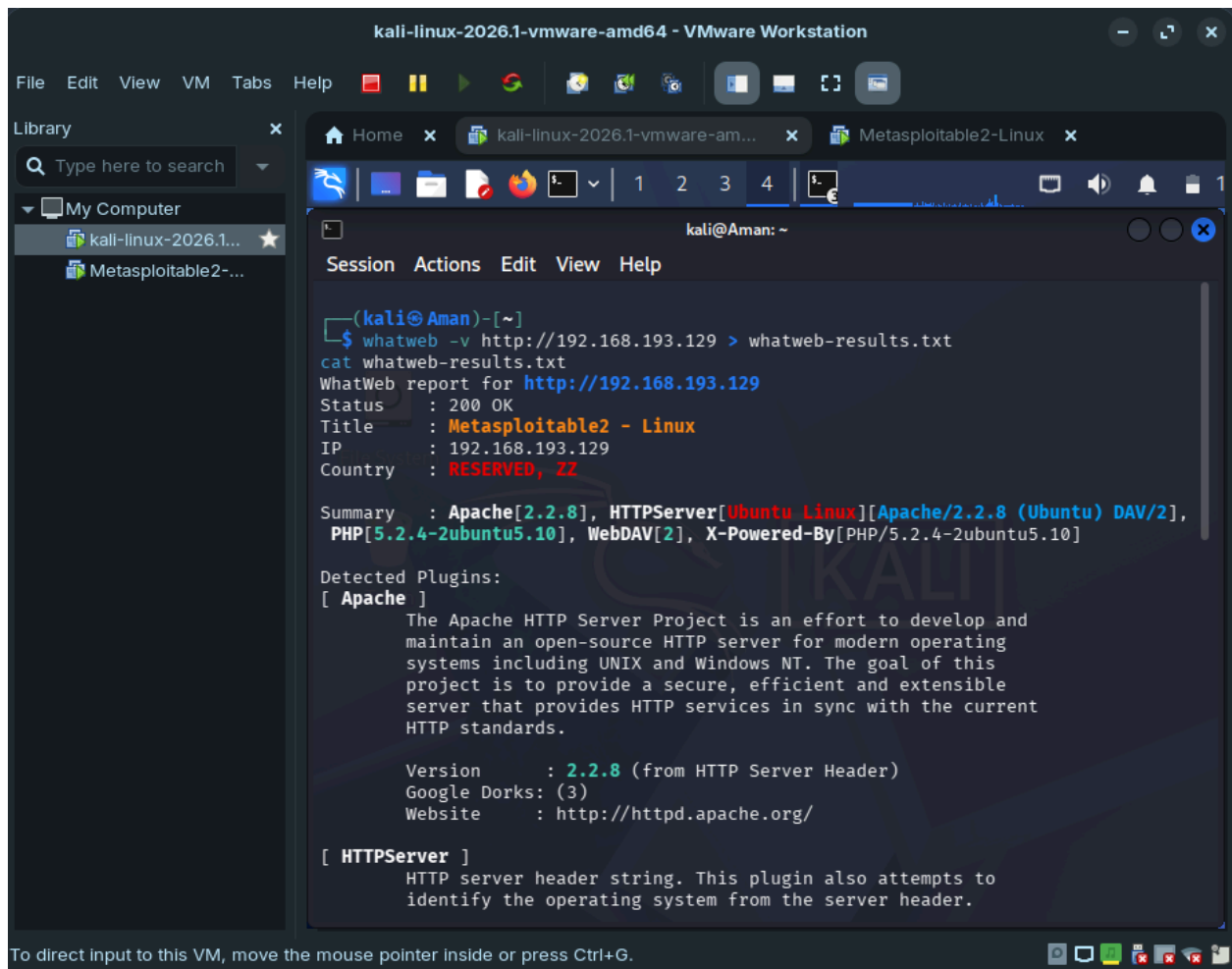
I used the command “whatweb --follow-redirect=always <http://192.168.193.129>” to

Redirect Occurred: No. The root web server returned a direct 200 OK response.

Observed Server Behavior: The server serves the main landing directory immediately rather than enforcing HTTP-to-HTTPS upgrades or automatic path redirections (e.g., 301 Moved Permanently or 302 Found).

Fingerprinted Destination: Identical to the initial root landing page (Title: Metasploitable2 - Linux, Apache 2.2.8, PHP 5.2.4-2ubuntu5.10).

Step 30: Save results



The screenshot shows a Kali Linux virtual machine running in VMware Workstation. The terminal window displays the output of the 'whatweb' command, which has been saved to a file named 'whatweb-results.txt'. The output shows a successful connection to the target IP address (192.168.193.129) and identifies the web server as Apache 2.2.8 running on Ubuntu Linux. The terminal also lists detected plugins, including Apache and HTTPServer, and provides details about the server's version and configuration.

```
(kali@Aman)-[~]
$ whatweb -v http://192.168.193.129 > whatweb-results.txt
cat whatweb-results.txt
WhatWeb report for http://192.168.193.129
Status      : 200 OK
Title       : Metasploitable2 - Linux
IP          : 192.168.193.129
Country     : RESERVED, ZZ

Summary     : Apache[2.2.8], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2],
             PHP[5.2.4-2ubuntu5.10], WebDAV[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]

Detected Plugins:
[ Apache ]
The Apache HTTP Server Project is an effort to develop and
maintain an open-source HTTP server for modern operating
systems including UNIX and Windows NT. The goal of this
project is to provide a secure, efficient and extensible
server that provides HTTP services in sync with the current
HTTP standards.

Version     : 2.2.8 (from HTTP Server Header)
Google Dorks: (3)
Website     : http://httpd.apache.org/

[ HTTPServer ]
HTTP server header string. This plugin also attempts to
identify the operating system from the server header.
```

Fig. 5J

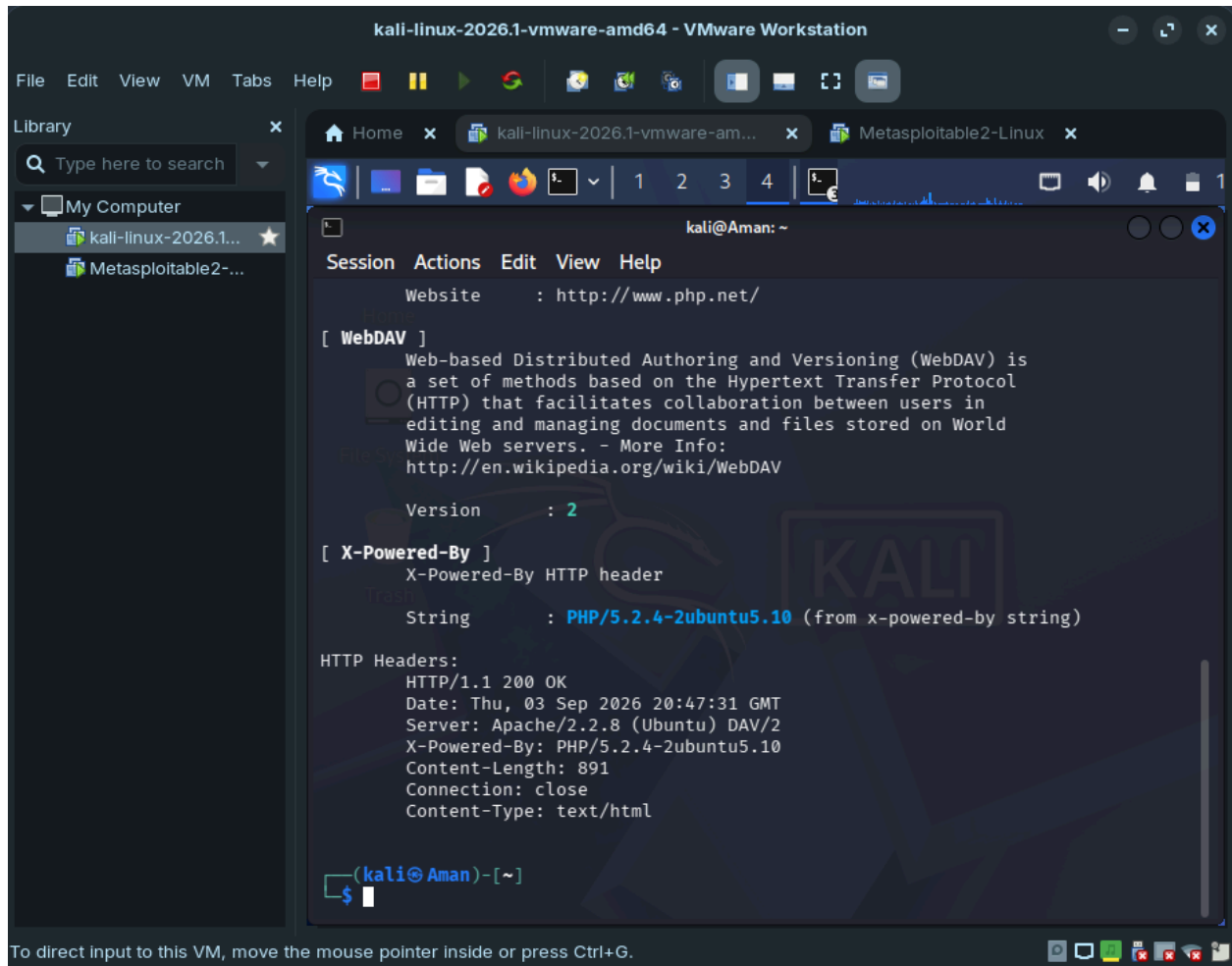


Fig 5K

I used the command “`whatweb -v http://<target-ip> > whatweb-results.txt, cat whatweb-results.txt`” to redirect the output to a text file.

Part 8 - Student Questions

1. What is reconnaissance?

Reconnaissance is the initial phase of a security assessment or penetration test where an auditor gathers information about a target system, network, or application. It involves mapping out active hosts, open network ports, running services, operating systems, and underlying technology stacks to establish a profile of the target's attack surface before executing any targeted actions.

2. What is the difference between host discovery and port scanning?

Host Discovery: Identifies which IP addresses are active ("alive") on a network segment (e.g., using ICMP echo requests, ARP requests, or TCP SYN/ACK probes to live hosts).

Port Scanning: Probes specific transport-layer ports (TCP/UDP) on a host already known to be active to determine which network services or applications are listening for connections.

3. Explain open, closed, and filtered.

Open: is when an application or service is actively accepting connections on the port

Closed: is when host is reachable, but no application is listening on the port

Filtered: is when Nmap cannot determine whether the port is open or closed because network security controls

4. What does -sV do?

The -sV flag enables Service Version Detection

5. What does -O do?

The -O flag enables Operating System Detection.

6. Explain what -A combines and why it is noisier.

The -A flag enables aggressive scan option

7. What does -p- mean?

-p- is a shorthand directive that instructs Nmap to scan all ports

8. Why can UDP scans take longer?

UDP is a connectionless protocol. When probing a closed UDP port, the target typically sends back an ICMP Port Unreachable error message; however, operating systems actively rate-limit ICMP responses (for example, Linux limits ICMP responses to 1 per second).

9. What does -sC do?

The -sC flag runs Nmap's default script set

10. What information did the HTTP title/headers scripts reveal?

The HTTP scripts revealed the HTTP title.

11. What did SMB/SSH/FTP enumeration add?

FTP Enumeration : Returned the unparsed initial banner string `

SSH Enumeration : Extracted public host key types

SMB Enumeration : Identified that the host exclusively supports the legacy.

12. What is the difference between Nmap service detection and WhatWeb fingerprinting?

Nmap Service Detection works primarily at the transport and network layer (TCP/UDP) across all network protocols to identify daemon binaries and versions using raw protocol response probes.

While WhatWeb Fingerprinting functions strictly at the application layer (HTTP/HTTPS) using specialized web plugins to identify content management systems, web frameworks, embedded JavaScript libraries, analytics tracking code, and structural web components that standard port scanners miss.

13. How did WhatWeb levels 1, 3 and 4 differ?

Level 1 : Makes a single HTTP request per URL and passively analyzes response headers, HTML titles, and meta tags without recursive crawling.

Level 3 : Recursively follows links, extracts embedded targets, executes secondary GET requests to known application paths, and triggers active plugin heuristics.

Level 4 : Conducts exhaustive, deep recursive crawling, submits forms, attempts brute-force path checks, and sends potentially intrusive requests to exhaustively map every web component (generating maximum traffic).

14. Why must aggressive scanning/fingerprinting remain inside the authorised lab?

Performing aggressive tools strictly within an isolated lab environment ensures compliance with legal scopes and ethical boundaries.